Diss cleaning

SCL

2026-08-08

```
library(pwr)
library(sjstats)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tibble)
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.3
```

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ forcats   1.0.1     ✓ readr     2.1.6
## ✓ ggplot2   4.0.2     ✓ stringr   1.5.2
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.2
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(here)
```

```
## here() starts at C:/Users/slewi/OneDrive/Documents/GitHub/Perception-of-researcher-roles
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
##
## The following object is masked from 'package:sjstats':
##
##     phi
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:psych':
##
##     logit
##
## The following object is masked from 'package:purrr':
##
##     some
##
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(effectsize)
```

```
##
## Attaching package: 'effectsize'
##
## The following object is masked from 'package:psych':
##
##     phi
##
## The following objects are masked from 'package:sjstats':
##
##     cohens_f, cramers_v, phi
```

```
library(broom)
```

```
##
## Attaching package: 'broom'
##
## The following object is masked from 'package:sjstats':
##
##     bootstrap
```

```
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
library(lattice)
library(nlme)
```

```
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```
library(lubridate)
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
##
## Attaching package: 'lme4'
##
## The following object is masked from 'package:nlme':
##
##     lmList
```

```
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.5.3
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```
style_table = function(kbl, footnote_text = NULL, header_above = NULL) {
  if (knitr::is_html_output()) {
    kbl = kableExtra::kable_styling(kbl, full_width = FALSE,
                                    bootstrap_options = "condensed")
    if (!is.null(header_above)) {
      kbl = kableExtra::add_header_above(kbl, header_above)
    }
    if (!is.null(footnote_text)) {
      kbl = kableExtra::footnote(kbl, general = footnote_text,
                                 general_title = "Note.",
                                 footnote_as_chunk = TRUE)
    }
    kbl
  } else {
    if (!is.null(footnote_text)) {
      cat(paste0("\n\n*Note.* ", footnote_text, "\n\n"))
    }
    kbl
  }
}
```

LOAD DATA

```
library(here)
data_path = here::here("Diss_full.csv")
diss.full =  read.csv(data_path)
```

LET'S START CLEANING

```
#make all the columns lowercase to help with coding
names(diss.full) = tolower(names(diss.full))
names(diss.full) =  iconv(names(diss.full), from = "", to = "UTF-8", sub = "")
names(diss.full) =  gsub("consent.*", "consent", names(diss.full))
```

Okay data is loaded. Now, we will clean it by the guidelines of the pre-reg.

Participants will be excluded if they: 1) do not consent or 2) complete the study in less than seven minutes or 3) exhibit no variability in their responses (e.g., selecting all 5s or all 3s) across all Integrity and likeability items, and 4) fail to complete the Integrity and likeability ratings for at least one vignettes.

An attention check will be used to assess whether participants paid attention during the experiment by requiring participants to select a specific response to the question (i.e., “Select the option “somewhat disagree” if you are paying attention.”). We will do an exploratory analysis to determine test if excluding those who fail the attention impacts the data.

```
# Initial data count
initial_count =  nrow(diss.full)
print(initial_count)
```

```
## [1] 675
```

```
# Clean recordeddate
diss.full$recordeddate =  str_squish(as.character(diss.full$recordeddate))
diss.full$created_date = mdy_hm(diss.full$recordeddate)

# Filters by cutoff date
cutoff_date =  as.Date("2026-01-27")
filtered_by_date = diss.full %>%
  filter(!is.na(created_date) & created_date >= cutoff_date)

count_by_date =  nrow(filtered_by_date)
print(count_by_date)
```

```
## [1] 675
```

```
# Filter by consent
filtered_by_consent = filtered_by_date %>%
  filter(consent == "Yes, I consent")

count_by_consent =  nrow(filtered_by_consent)

# Filter by duration (exclude under 420 seconds)
filtered_by_duration = filtered_by_consent %>%
  filter(!is.na(duration..in.seconds.) & duration..in.seconds. >= 420)

count_by_duration = nrow(filtered_by_duration)

# Exclude participants under 18
filtered_by_age = filtered_by_duration %>%
  filter(is.na(age) | age >= 18)

count_by_age = nrow(filtered_by_age)


# Exclude no-variability: participants who gave identical responses across
# all Integrity and likeability items — uses apply() to avoid rowwise() issues
Integrity_like_cols = c("khon","ksin","keth","klike","kapp",
                    "shon","ssin","seth","slike","sapp",
                    "hhon","hsin","heth","hlike","happ",
                    "fhon","fsin","feth","flike","fapp",
                    "xhon","xsin","xeth","xlike","xapp",
                    "yhon","ysin","yeth","ylike","yapp")

tl_matrix = filtered_by_age %>%
  select(all_of(Integrity_like_cols)) %>%
  mutate(across(everything(), as.numeric))
```

```
## Warning: There were 30 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(everything(), as.numeric)`.
## Caused by warning:
## ! NAs introduced by coercion
## i Run `dplyr::last_dplyr_warnings()` to see the 29 remaining warnings.
```

```
has_variability = apply(tl_matrix, 1, function(row) n_distinct(na.omit(row)) > 1)
filtered_by_variability = filtered_by_age[has_variability, ]
count_by_variability = nrow(filtered_by_variability)

# Exclude attention check failures (attention1 == 2 is correct response)
filtered_by_attention = filtered_by_variability %>%
  filter(attention1 == 2)
count_by_attention = nrow(filtered_by_attention)
# Final count
final_count = nrow(filtered_by_age)

# Summary table
count_summary = data.frame(
  Step = c(
    "Initial",
    "After Date Filter",
    "After Consent Filter",
    "After Duration Filter",
    "After Age Filter",
    "Final"
  ),
  Count = c(
    initial_count,
    count_by_date,
    count_by_consent,
    count_by_duration,
    count_by_age,
    final_count
  )
)

print(count_summary)
```

```
##                   Step Count
## 1              Initial   675
## 2     After Date Filter   675
## 3  After Consent Filter   675
## 4 After Duration Filter   455
## 5      After Age Filter   455
## 6                Final   455
```

```
# Print counts
cat("Total rows in diss.full:", initial_count, "\n")
```

```
## Total rows in diss.full: 675
```

```
cat("Rows after filtering by date:", count_by_date, "\n")
```

```
## Rows after filtering by date: 675
```

```
cat("Rows after filtering by consent:", count_by_consent, "\n")
```

```
## Rows after filtering by consent: 675
```

```
cat("Rows after filtering by duration:", count_by_duration, "\n")
```

```
## Rows after filtering by duration: 455
```

```
cat("Rows after excluding participants under 18:", count_by_age, "\n")
```

```
## Rows after excluding participants under 18: 455
```

```
cat("Final rows:", final_count, "\n")
```

```
## Final rows: 455
```

okay now we filter by ege

```
attr_dat = filtered_by_age %>%
  select(
    responseid,
    minsmart, minbetter, mincreative, minrather, minshould, minregard, mintyp, minkeep, minget,
    modsmart, modbett, modcreat, modrath, modshou, modreg, modtyp, modkeep, modget,
    majsmart, majbet, majcreat, majrath, majshould, majreg, majtyp, majkeep, majget
  )
```

Pivot the dataset

```
attr_long = attr_dat %>%
  pivot_longer(
    cols = -responseid,
    names_to = "raw_item",
    values_to = "choice_raw"
  )
```

find the correct columns to attribute to the variables

```
attr_long = attr_long %>%
  mutate(
    severity = case_when(
      str_detect(raw_item, "^min") ~ "minor",
      str_detect(raw_item, "^mod") ~ "moderate",
      str_detect(raw_item, "^maj") ~ "major",
      TRUE ~ NA_character_
    ),
    item_stub = str_remove(raw_item, "^(min|mod|maj)"),
    attribute = case_when(
      item_stub %in% c("smart") ~ "smarter",
      item_stub %in% c("better", "bett") ~ "better_researcher",
      item_stub %in% c("creative", "creat") ~ "creative",
      item_stub %in% c("rather", "rath") ~ "rather_be",
      item_stub %in% c("should", "shou") ~ "should_be",
      item_stub %in% c("regard", "reg") ~ "celebrated",
      item_stub %in% c("typ") ~ "typical",
      item_stub %in% c("keep") ~ "job_security",
      item_stub %in% c("get") ~ "job_outcome",
      TRUE ~ NA_character_
    )
  )
```

Check to make sure it looks correct and get a count

```
head(attr_long, 20)
```

```
## # A tibble: 20 × 6
##    responseid        raw_item    choice_raw         severity item_stub attribute
##    <chr>             <chr>       <chr>              <chr>    <chr>     <chr>
##  1 R_1kWnphloUO7A7Tj minsmart    neither reported … minor    smart     smarter
##  2 R_1kWnphloUO7A7Tj minbetter   reporting researc… minor    better    better_r…
##  3 R_1kWnphloUO7A7Tj mincreative reporting researc… minor    creative  creative
##  4 R_1kWnphloUO7A7Tj minrather   reporting researc… minor    rather    rather_be
##  5 R_1kWnphloUO7A7Tj minshould   reporting researc… minor    should    should_be
##  6 R_1kWnphloUO7A7Tj minregard   reporting researc… minor    regard    celebrat…
##  7 R_1kWnphloUO7A7Tj mintyp      neither reported … minor    typ       typical
##  8 R_1kWnphloUO7A7Tj minkeep     reporting researc… minor    keep      job_secu…
##  9 R_1kWnphloUO7A7Tj minget      reporting researc… minor    get       job_outc…
## 10 R_1kWnphloUO7A7Tj modsmart    reporting researc… moderate smart     smarter
## 11 R_1kWnphloUO7A7Tj modbett     reporting researc… moderate bett      better_r…
## 12 R_1kWnphloUO7A7Tj modcreat    neither reported … moderate creat     creative
## 13 R_1kWnphloUO7A7Tj modrath     reporting researc… moderate rath      rather_be
## 14 R_1kWnphloUO7A7Tj modshou     reporting researc… moderate shou      should_be
## 15 R_1kWnphloUO7A7Tj modreg      reporting researc… moderate reg       celebrat…
## 16 R_1kWnphloUO7A7Tj modtyp      neither reported … moderate typ       typical
## 17 R_1kWnphloUO7A7Tj modkeep     reporting researc… moderate keep      job_secu…
## 18 R_1kWnphloUO7A7Tj modget      reporting researc… moderate get       job_outc…
## 19 R_1kWnphloUO7A7Tj majsmart    neither reported … major    smart     smarter
## 20 R_1kWnphloUO7A7Tj majbet      reporting researc… major    bet       <NA>
```

```
table(attr_long$severity, useNA = "ifany")
```

```
##
##    major    minor moderate
##     4095     4095     4095
```

```
table(attr_long$attribute, useNA = "ifany")
```

```
##
## better_researcher        celebrated          creative       job_outcome
##               910              1365              1365              1365
##      job_security         rather_be         should_be           smarter
##              1365              1365              1365              1365
##           typical              <NA>
##              1365               455
```

set roles

```
attr_long2 = attr_long %>%
  tidyr::crossing(researcher = c("investigator", "inspector"))
```

figure out how many rows for the data

```
head(attr_long2, 20)
```

```
## # A tibble: 20 × 7
##    responseid        raw_item choice_raw severity item_stub attribute researcher
##    <chr>             <chr>    <chr>      <chr>    <chr>     <chr>     <chr>
##  1 R_100RpLnF3upbBKT majbet   reporting… major    bet       <NA>      inspector
##  2 R_100RpLnF3upbBKT majbet   reporting… major    bet       <NA>      investiga…
##  3 R_100RpLnF3upbBKT majcreat neither r… major    creat     creative  inspector
##  4 R_100RpLnF3upbBKT majcreat neither r… major    creat     creative  investiga…
##  5 R_100RpLnF3upbBKT majget   reporting… major    get       job_outc… inspector
##  6 R_100RpLnF3upbBKT majget   reporting… major    get       job_outc… investiga…
##  7 R_100RpLnF3upbBKT majkeep  reporting… major    keep      job_secu… inspector
##  8 R_100RpLnF3upbBKT majkeep  reporting… major    keep      job_secu… investiga…
##  9 R_100RpLnF3upbBKT majrath  reporting… major    rath      rather_be inspector
## 10 R_100RpLnF3upbBKT majrath  reporting… major    rath      rather_be investiga…
## 11 R_100RpLnF3upbBKT majreg   reporting… major    reg       celebrat… inspector
## 12 R_100RpLnF3upbBKT majreg   reporting… major    reg       celebrat… investiga…
## 13 R_100RpLnF3upbBKT majshou… reporting… major    should    should_be inspector
## 14 R_100RpLnF3upbBKT majshou… reporting… major    should    should_be investiga…
## 15 R_100RpLnF3upbBKT majsmart reporting… major    smart     smarter   inspector
## 16 R_100RpLnF3upbBKT majsmart reporting… major    smart     smarter   investiga…
## 17 R_100RpLnF3upbBKT majtyp   reported … major    typ       typical   inspector
## 18 R_100RpLnF3upbBKT majtyp   reported … major    typ       typical   investiga…
## 19 R_100RpLnF3upbBKT minbett… neither r… minor    better    better_r… inspector
## 20 R_100RpLnF3upbBKT minbett… neither r… minor    better    better_r… investiga…
```

```
dim(attr_long)
```

```
## [1] 12285     6
```

```
dim(attr_long2)
```

```
## [1] 24570     7
```

find the correct columns and attruibute them to role type

```
attr_long2 = attr_long2 %>%
  mutate(
    chosen = case_when(
      str_detect(choice_raw, "^reported researcher") & researcher == "investigator" ~ 1,
      str_detect(choice_raw, "^reporting researcher") & researcher == "inspector" ~ 1,
      str_detect(choice_raw, "^neither") ~ 0,
      TRUE ~ 0
    )
  )
```

Check to make sure it looks correct and get a count

```
head(attr_long2, 20)
```

```
## # A tibble: 20 × 8
##    responseid raw_item choice_raw severity item_stub attribute researcher chosen
##    <chr>      <chr>    <chr>      <chr>    <chr>     <chr>     <chr>       <dbl>
##  1 R_100RpLn… majbet   reporting… major    bet       <NA>      inspector       1
##  2 R_100RpLn… majbet   reporting… major    bet       <NA>      investiga…      0
##  3 R_100RpLn… majcreat neither r… major    creat     creative  inspector       0
##  4 R_100RpLn… majcreat neither r… major    creat     creative  investiga…      0
##  5 R_100RpLn… majget   reporting… major    get       job_outc… inspector       1
##  6 R_100RpLn… majget   reporting… major    get       job_outc… investiga…      0
##  7 R_100RpLn… majkeep  reporting… major    keep      job_secu… inspector       1
##  8 R_100RpLn… majkeep  reporting… major    keep      job_secu… investiga…      0
##  9 R_100RpLn… majrath  reporting… major    rath      rather_be inspector       1
## 10 R_100RpLn… majrath  reporting… major    rath      rather_be investiga…      0
## 11 R_100RpLn… majreg   reporting… major    reg       celebrat… inspector       1
## 12 R_100RpLn… majreg   reporting… major    reg       celebrat… investiga…      0
## 13 R_100RpLn… majshou… reporting… major    should    should_be inspector       1
## 14 R_100RpLn… majshou… reporting… major    should    should_be investiga…      0
## 15 R_100RpLn… majsmart reporting… major    smart     smarter   inspector       1
## 16 R_100RpLn… majsmart reporting… major    smart     smarter   investiga…      0
## 17 R_100RpLn… majtyp   reported … major    typ       typical   inspector       0
## 18 R_100RpLn… majtyp   reported … major    typ       typical   investiga…      1
## 19 R_100RpLn… minbett… neither r… minor    better    better_r… inspector       0
## 20 R_100RpLn… minbett… neither r… minor    better    better_r… investiga…      0
```

```
table(attr_long2$chosen, useNA = "ifany")
```

```
##
##     0     1
## 15303  9267
```

```
table(attr_long2$researcher, attr_long2$chosen)
```

```
##
##                    0     1
##   inspector     4731  7554
##   investigator 10572  1713
```

set up the data for the models

```
attr_model_dat = attr_long2 %>%
  mutate(
    researcher_ec = if_else(researcher == "inspector", 1, -1),
    severity_num = case_when(
      severity == "minor" ~ -1,
      severity == "moderate" ~ 0,
      severity == "major" ~ 1
    ),
    responseid = factor(responseid),
    researcher = factor(researcher, levels = c("investigator", "inspector")),
    severity = factor(severity, levels = c("minor", "moderate", "major")),
    attribute = factor(
      attribute,
      levels = c(
        "smarter",
        "better_researcher",
        "creative",
        "rather_be",
        "should_be",
        "celebrated",
        "typical",
        "job_security",
        "job_outcome"
      )
    ),
    vignette_id = factor(severity)
  )
```

create some model tables

```
str(attr_model_dat)
```

```
## tibble [24,570 × 11] (S3: tbl_df/tbl/data.frame)
##  $ responseid   : Factor w/ 455 levels "R_100RpLnF3upbBKT",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ raw_item     : chr [1:24570] "majbet" "majbet" "majcreat" "majcreat" ...
##  $ choice_raw   : chr [1:24570] "reporting researcher (Y)" "reporting researcher (Y)" "neither reported (X) nor reporting
researcher (Y)" "neither reported (X) nor reporting researcher (Y)" ...
##  $ severity     : Factor w/ 3 levels "minor","moderate",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ item_stub    : chr [1:24570] "bet" "bet" "creat" "creat" ...
##  $ attribute    : Factor w/ 9 levels "smarter","better_researcher",..: NA NA 3 3 9 9 8 8 4 4 ...
##  $ researcher   : Factor w/ 2 levels "investigator",..: 2 1 2 1 2 1 2 1 2 1 ...
##  $ chosen       : num [1:24570] 1 0 0 0 1 0 1 0 1 0 ...
##  $ researcher_ec: num [1:24570] 1 -1 1 -1 1 -1 1 -1 1 -1 ...
##  $ severity_num : num [1:24570] 1 1 1 1 1 1 1 1 1 1 ...
##  $ vignette_id  : Factor w/ 3 levels "minor","moderate",..: 3 3 3 3 3 3 3 3 3 3 ...
```

```
table(attr_model_dat$researcher)
```

```
##
## investigator    inspector
##        12285        12285
```

```
table(attr_model_dat$severity)
```

```
##
##    minor moderate    major
##     8190     8190     8190
```

```
table(attr_model_dat$attribute)
```

```
##
##           smarter better_researcher          creative         rather_be
##              2730              1820              2730              2730
##         should_be        celebrated           typical      job_security
##              2730              2730              2730              2730
##       job_outcome
##              2730
```

```
table(attr_model_dat$researcher_ec)
```

```
##
##    -1     1
## 12285 12285
```

```
table(attr_model_dat$severity_num)
```

```
##
##   -1    0    1
## 8190 8190 8190
```

Run our first model in the model build for hypo 1

```
A1 = glmer(
  chosen ~ researcher_ec + attribute +
    (1 | responseid) + (1 | vignette_id),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa")
)
```

summarize our baseline

```
# chosen = 0 (not selected), 1 (selected)
summary(A1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: chosen ~ researcher_ec + attribute + (1 | responseid) + (1 |
##     vignette_id)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   24538.9   24635.7  -12257.4   24514.9     23648
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0024 -0.5132 -0.3430  0.6463  7.9218
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  responseid  (Intercept) 0.418855 0.64719
##  vignette_id (Intercept) 0.002816 0.05306
## Number of obs: 23660, groups:  responseid, 455; vignette_id, 3
##
## Fixed effects:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -1.20546    0.06522 -18.482  < 2e-16 ***
## researcher_ec               1.20467    0.01700  70.848  < 2e-16 ***
## attributebetter_researcher  0.53365    0.07540   7.078 1.47e-12 ***
## attributecreative          -0.14470    0.06855  -2.111   0.0348 *
## attributerather_be          0.53692    0.06702   8.011 1.13e-15 ***
## attributeshould_be          0.64749    0.06694   9.673  < 2e-16 ***
## attributecelebrated         0.69276    0.06692  10.352  < 2e-16 ***
## attributetypical            0.46043    0.06710   6.862 6.79e-12 ***
## attributejob_security       0.77005    0.06690  11.511  < 2e-16 ***
## attributejob_outcome        0.73360    0.06691  10.965  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ attrbtb_ attrbtcr attrbtr_ attrbts_ attrbtcl attrbtt
## researchr_c -0.131
## attrbtbttr_ -0.477  0.046
## attribtcrtv -0.516 -0.011  0.447
## attrbtrthr_ -0.537  0.053  0.461    0.503
## attrbtshld_ -0.539  0.066  0.462    0.504    0.520
## attrbtclbrt -0.540  0.071  0.462    0.504    0.520    0.522
## attrbttypcl -0.535  0.044  0.460    0.503    0.517    0.519    0.519
## attrbtjb_sc -0.542  0.080  0.463    0.504    0.521    0.523    0.523    0.520
## attrbtjb_tc -0.541  0.076  0.463    0.504    0.521    0.522    0.523    0.519
##             attrbtjb_s
## researchr_c
## attrbtbttr_
## attribtcrtv
## attrbtrthr_
## attrbtshld_
## attrbtclbrt
## attrbttypcl
## attrbtjb_sc
## attrbtjb_tc  0.524
```

run model two

```
# A2 tests whether choice differs by severity, controlling for attribute
# chosen = 0 (not selected), 1 (selected)

A2 = glmer(
  chosen ~ attribute + severity_num +
    (1 | responseid),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
)

summary(A2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: chosen ~ attribute + severity_num + (1 | responseid)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   30664.2   30753.0  -15321.1   30642.2     23649
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.0647 -0.8057 -0.6120  1.1111  2.8606
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  responseid (Intercept) 0.2131   0.4617
## Number of obs: 23660, groups:  responseid, 455
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.89499    0.04758 -18.809  < 2e-16 ***
## attributebetter_researcher  0.41379    0.06524   6.342 2.26e-10 ***
## attributecreative          -0.11272    0.06054  -1.862  0.06263 .
## attributerather_be          0.40517    0.05813   6.970 3.17e-12 ***
## attributeshould_be          0.48661    0.05792   8.401  < 2e-16 ***
## attributecelebrated         0.51981    0.05785   8.986  < 2e-16 ***
## attributetypical            0.34851    0.05830   5.978 2.26e-09 ***
## attributejob_security       0.57635    0.05774   9.982  < 2e-16 ***
## attributejob_outcome        0.54971    0.05779   9.512  < 2e-16 ***
## severity_num                0.04607    0.01719   2.680  0.00736 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) attrbtb_ attrbtcr attrbtr_ attrbts_ attrbtcl attrbtt
## attrbtbttr_ -0.578
## attribtcrtv -0.622  0.453
## attrbtrthr_ -0.648  0.473    0.509
## attrbtshld_ -0.650  0.474    0.511    0.532
## attrbtclbrt -0.651  0.475    0.511    0.533    0.535
## attrbttypcl -0.646  0.471    0.507    0.529    0.530    0.531
## attrbtjb_sc -0.652  0.476    0.512    0.534    0.536    0.536    0.532
## attrbtjb_tc -0.652  0.475    0.512    0.533    0.535    0.536    0.532
## severity_nm -0.004  0.134    0.000    0.002    0.002    0.002    0.001
##             attrbtjb_s attrbtjb_t
## attrbtbttr_
## attribtcrtv
## attrbtrthr_
## attrbtshld_
## attrbtclbrt
## attrbttypcl
## attrbtjb_sc
## attrbtjb_tc  0.537
## severity_nm  0.002      0.002
```

run model three

```
# A3: Tests whether choice differs by BOTH researcher and severity,
# while controlling for attribute type

# DV: chosen (0 = not selected, 1 = selected)

A3 = glmer(
  chosen ~ researcher_ec + attribute + severity_num +
    (1 | responseid),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
)

summary(A3)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: chosen ~ researcher_ec + attribute + severity_num + (1 | responseid)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   24535.2   24632.0  -12255.6   24511.2     23648
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0312 -0.5124 -0.3415  0.6475  7.9244
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  responseid (Intercept) 0.4182   0.6467
## Number of obs: 23660, groups:  responseid, 455
##
## Fixed effects:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -1.20543    0.05757 -20.938  < 2e-16 ***
## researcher_ec               1.20464    0.01701  70.835  < 2e-16 ***
## attributebetter_researcher  0.54830    0.07537   7.274 3.48e-13 ***
## attributecreative          -0.14470    0.06855  -2.111  0.03479 *
## attributerather_be          0.53691    0.06702   8.011 1.14e-15 ***
## attributeshould_be          0.64748    0.06694   9.672  < 2e-16 ***
## attributecelebrated         0.69275    0.06692  10.351  < 2e-16 ***
## attributetypical            0.46042    0.06710   6.862 6.81e-12 ***
## attributejob_security       0.77004    0.06690  11.510  < 2e-16 ***
## attributejob_outcome        0.73358    0.06691  10.964  < 2e-16 ***
## severity_num                0.06187    0.01998   3.097  0.00196 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ attrbtb_ attrbtcr attrbtr_ attrbts_ attrbtcl attrbtt
## researchr_c -0.148
## attrbtbttr_ -0.542  0.048
## attribtcrtv -0.585 -0.011  0.447
## attrbtrthr_ -0.608  0.053  0.461    0.503
## attrbtshld_ -0.611  0.066  0.462    0.504    0.520
## attrbtclbrt -0.612  0.071  0.463    0.504    0.520    0.522
## attrbttypcl -0.606  0.044  0.460    0.503    0.517    0.519    0.519
## attrbtjb_sc -0.614  0.080  0.463    0.504    0.521    0.523    0.523    0.520
## attrbtjb_tc -0.613  0.076  0.463    0.504    0.521    0.522    0.523    0.519
## severity_nm -0.006  0.022  0.135   -0.001    0.002    0.003    0.003    0.002
##             attrbtjb_s attrbtjb_t
## researchr_c
## attrbtbttr_
## attribtcrtv
## attrbtrthr_
## attrbtshld_
## attrbtclbrt
## attrbttypcl
## attrbtjb_sc
## attrbtjb_tc  0.524
## severity_nm  0.003      0.003
```

run model four

```
# A4: Tests whether the effect of researcher depends on severity,
# while controlling for attribute type

# DV: chosen (0 = not selected, 1 = selected)

A4 = glmer(
  chosen ~ researcher_ec * severity_num + attribute +
    (1 | responseid),
  data = attr_model_dat,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
)

summary(A4)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: chosen ~ researcher_ec * severity_num + attribute + (1 | responseid)
##    Data: attr_model_dat
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   24164.9   24269.9  -12069.5   24138.9     23647
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.3906 -0.5219 -0.3119  0.6375  8.1662
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  responseid (Intercept) 0.436    0.6603
## Number of obs: 23660, groups:  responseid, 455
##
## Fixed effects:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -1.24261    0.05854 -21.228  < 2e-16 ***
## researcher_ec               1.25005    0.01771  70.580  < 2e-16 ***
## severity_num               -0.05793    0.02160  -2.682  0.00733 **
## attributebetter_researcher  0.55980    0.07439   7.525 5.27e-14 ***
## attributecreative          -0.14787    0.06931  -2.134  0.03288 *
## attributerather_be          0.54939    0.06781   8.102 5.40e-16 ***
## attributeshould_be          0.66269    0.06774   9.783  < 2e-16 ***
## attributecelebrated         0.70909    0.06773  10.470  < 2e-16 ***
## attributetypical            0.47104    0.06788   6.939 3.94e-12 ***
## attributejob_security       0.78835    0.06771  11.642  < 2e-16 ***
## attributejob_outcome        0.75097    0.06772  11.090  < 2e-16 ***
## researcher_ec:severity_num  0.40581    0.02141  18.951  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) rsrch_ svrty_ attrbtb_ attrbtcr attrbtr_ attrbts_ attrbtcl
## researchr_c -0.162
## severity_nm  0.053 -0.151
## attrbtbttr_ -0.549  0.050  0.113
## attribtcrtv -0.582 -0.012 -0.002  0.458
## attrbtrthr_ -0.605  0.054  0.004  0.474    0.503
## attrbtshld_ -0.608  0.067  0.004  0.475    0.503    0.519
## attrbtclbrt -0.609  0.072  0.004  0.476    0.503    0.520    0.521
## attrbttypcl -0.603  0.045  0.004  0.473    0.503    0.517    0.518    0.519
## attrbtjb_sc -0.611  0.081  0.004  0.476    0.503    0.520    0.522    0.523
## attrbtjb_tc -0.610  0.077  0.004  0.476    0.503    0.520    0.522    0.522
## rsrchr_c:s_ -0.065  0.237 -0.321  0.016   -0.004    0.015    0.018    0.019
##             attrbtt attrbtjb_s attrbtjb_t
## researchr_c
## severity_nm
## attrbtbttr_
## attribtcrtv
## attrbtrthr_
## attrbtshld_
## attrbtclbrt
## attrbttypcl
## attrbtjb_sc  0.519
## attrbtjb_tc  0.519   0.523
## rsrchr_c:s_  0.012   0.022      0.021
```

Integrity & LIKEABILITY ANALYSIS (RQ1, RQ2) + TRUENESS (RQ3)

Research Question 1: How is integrity influenced by role and QRP severity? • Hypothesis 1a: Inspectors will be rated higher in integrity than investigators. • Hypothesis 1b: The relative integrity of inspectors versus investigators will increase as QRP severity increases.

Research Question 2: How is likeability influenced by role and QRP severity? • Hypothesis 2a: Investigators will be rated more likable than inspectors. • Hypothesis 2b: At low QRP severity, investigators will be rated as more likeable than inspectors, but at high QRP severity, inspectors will be rated as more likeable than investigators.

Research Question 3: Is the perceived trueness of the finding influenced by QRP severity? • Hypothesis 3: Perceived trueness will decrease as QRP severity increases.

Research Questions 1–2: Integrity and Likeability Step-up MLMs will test Role, Severity, and Role × Severity interaction:

Model 1: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + v0k + μ0jk + εijk Model 2: Integrity/Likeabilityijk = γ000 + γ010 (Severityk) + v0k + μ0k + εijk Model 3: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + γ010 (Severityk) + v0k + μ0jk + εijk Model 4: Integrity/Likeabilityijk = γ000 + γ100 (Researcherj) + γ010 (Severityk) + γ110(Researcherj × Severityk) + v0k + μ0jk + εijk

Research Question 3: Trueness of Findings Model FT1: Truenessijk= γ000 + γ010 (Severityk) + v0k + μ0j + εijk

STEP 1: Build the long-format Integrity/likeability/trueness dataset

```
# Vignette metadata
# Each vignette pair: investigator (inv) and inspector (insp)
# Severity:minor = vignette K/S, moderate = H/F, major = X/Y
# Integrity items: hon, sin, eth  (honest, sincere, ethical)
# Likeability: like, app (likeable, approachable)
# Trueness (conf): investigator only, 1 item per vignette

vignette_map = list(
  list(vig_letter = "k", vig_id = "minor",    role = "investigator",
       hon = "khon", sin = "ksin",  eth = "keth",
       like = "klike",  app = "kapp",  conf = "kcon"),
  list(vig_letter = "s", vig_id = "minor",    role = "inspector",
       hon = "shon",    sin = "ssin",  eth = "seth",
       like = "slike",  app = "sapp",  conf = NA),
  list(vig_letter = "h", vig_id = "moderate", role = "investigator",
       hon = "hhon",    sin = "hsin",  eth = "heth",
       like = "hlike",  app = "happ",  conf = "hcon"),
  list(vig_letter = "f", vig_id = "moderate", role = "inspector",
       hon = "fhon",    sin = "fsin",  eth = "feth",
       like = "flike",  app = "fapp",  conf = NA),
  list(vig_letter = "x", vig_id = "major",    role = "investigator",
       hon = "xhon",   sin = "xsin",  eth = "xeth",
       like = "xlike",  app = "xapp",  conf = "xcon"),
  list(vig_letter = "y", vig_id = "major",    role = "inspector",
       hon = "yhon",    sin = "ysin",  eth = "yeth",
       like = "ylike",  app = "yapp",  conf = NA)
)
# vig_letter is the TRUE vignette identifier (6 distinct narrative vignettes:
# k/s/h/f/x/y). vig_id (severity) is NOT a valid vignette-level random effect
# on its own — it only has 3 levels and is perfectly confounded with
# severity_num, the key fixed effect. vig_letter fixes this: it captures
# idiosyncratic vignette-writing variance independent of role and severity.

#Helper: extract numeric value from labelled string
parse_num = function(x) as.numeric(str_extract(as.character(x), "^\\d"))

#Build one row per participant × vignette × role
tl_rows = bind_rows(lapply(vignette_map, function(v) {
  filtered_by_age %>%                          # use cleaned dataset
    transmute(
      responseid,
      vig_letter = v$vig_letter,
      vig_id   = v$vig_id,
      role     = v$role,
      hon_raw  = parse_num(.data[[v$hon]]),
      sin_raw  = parse_num(.data[[v$sin]]),
      eth_raw  = parse_num(.data[[v$eth]]),
      like_raw = parse_num(.data[[v$like]]),
      app_raw  = parse_num(.data[[v$app]]),
      conf_raw = if (!is.na(v$conf)) parse_num(.data[[v$conf]]) else NA_real_
    )
}))

#Composites & factors
tl_dat = tl_rows %>%
  mutate(
    Integrity      = rowMeans(cbind(hon_raw, sin_raw, eth_raw), na.rm = TRUE),
    likeability = rowMeans(cbind(like_raw, app_raw), na.rm = TRUE),
    severity   = factor(vig_id,   levels = c("minor", "moderate", "major")),
    role       = factor(role,     levels = c("investigator", "inspector")),
    responseid = factor(responseid),
    vignette_id = factor(vig_letter, levels = c("k","s","h","f","x","y")),
    # effect-coded predictors
    role_ec    = if_else(role == "inspector", 1, -1),
    severity_num = case_when(
      severity == "minor"    ~ -1,
      severity == "moderate" ~  0,
      severity == "major"    ~  1
    )
  )

glimpse(tl_dat)
```

```
## Rows: 2,730
## Columns: 16
## $ responseid   <fct> R_1kWnphloUO7A7Tj, R_5lX85aTYV0koeHL, R_1fs9EL5QZj1yBtB, …
## $ vig_letter   <chr> "k", "k", "k", "k", "k", "k", "k", "k", "k", "k", "k", "k…
## $ vig_id       <chr> "minor", "minor", "minor", "minor", "minor", "minor", "mi…
## $ role         <fct> investigator, investigator, investigator, investigator, i…
## $ hon_raw      <dbl> 3, 5, 2, 4, 3, 2, 3, 2, 5, 3, 2, 2, 2, 2, 4, 2, 4, 2, 4, …
## $ sin_raw      <dbl> 2, 5, 2, 4, 2, 4, 4, 2, 5, 4, 2, 4, 3, 2, 4, 3, 4, 3, 2, …
## $ eth_raw      <dbl> 1, 5, 3, 2, 4, 4, 2, 2, 5, 4, 1, 2, 2, 1, 4, 3, 3, 2, 2, …
## $ like_raw     <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 5, 2, 2, 4, 4, 3, 3, 3, 3, 3, 3, …
## $ app_raw      <dbl> 2, 4, 3, 5, 3, 4, 4, 4, 5, 4, 2, 4, 4, 4, 2, 4, 3, 3, 2, …
## $ conf_raw     <dbl> 3, 3, 3, 4, 4, 3, 3, 3, 4, 3, 2, 4, 3, 2, 3, 4, 3, 4, 2, …
## $ Integrity    <dbl> 2.000000, 5.000000, 2.333333, 3.333333, 3.000000, 3.33333…
## $ likeability  <dbl> 2.5, 3.5, 3.0, 4.0, 3.0, 3.5, 3.5, 3.5, 5.0, 3.0, 2.0, 4.…
## $ severity     <fct> minor, minor, minor, minor, minor, minor, minor, minor, m…
## $ vignette_id  <fct> k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, k, …
## $ role_ec      <dbl> -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -…
## $ severity_num <dbl> -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -…
```

Check structure and counts

```
table(tl_dat$role, tl_dat$severity)
```

```
##
##               minor moderate major
##   investigator   455      455   455
##   inspector      455      455   455
```

```
# Quick descriptives
tl_dat %>%
  group_by(role, severity) %>%
  summarise(
    n          = sum(!is.na(Integrity)),
    Integrity_M    = round(mean(Integrity, na.rm = TRUE), 2),
    Integrity_SD   = round(sd(Integrity,   na.rm = TRUE), 2),
    like_M     = round(mean(likeability, na.rm = TRUE), 2),
    like_SD    = round(sd(likeability,   na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  kable(caption = "Descriptives: Integrity & Likeability by Role × Severity") %>%
  style_table()
```

Descriptives: Integrity & Likeability by Role × Severity

role	severity	n	Integrity_M	Integrity_SD	like_M	like_SD
investigator	minor	446	2.79	0.86	3.03	0.77
investigator	moderate	447	1.69	0.73	2.36	0.81
investigator	major	447	1.39	0.70	2.02	0.84
inspector	minor	445	4.17	0.83	3.59	0.94
inspector	moderate	447	4.35	0.73	3.87	0.86
inspector	major	446	4.40	0.73	3.94	0.86

STEP 2: Reliability & validation checks

```
library(psych)

#Integrity alpha per role × severity (using the 3 items)
Integrity_items_inv = filtered_by_age %>%
  transmute(
    min_hon = parse_num(khon), min_sin = parse_num(ksin),   min_eth = parse_num(keth),
    mod_hon = parse_num(hhon),    mod_sin = parse_num(hsin),   mod_eth = parse_num(heth),
    maj_hon = parse_num(xhon),   maj_sin = parse_num(xsin),   maj_eth = parse_num(xeth)
  )

Integrity_items_insp = filtered_by_age %>%
  transmute(
    min_hon = parse_num(shon),  min_sin = parse_num(ssin),  min_eth = parse_num(seth),
    mod_hon = parse_num(fhon),  mod_sin = parse_num(fsin),  mod_eth = parse_num(feth),
    maj_hon = parse_num(yhon),  maj_sin = parse_num(ysin),  maj_eth = parse_num(yeth)
  )

psych::alpha(Integrity_items_inv,  check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.7740442 0.7812991
```

```
psych::alpha(Integrity_items_insp, check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.8659431 0.8674587
```

```
#Likeability alpha
like_items_inv = filtered_by_age %>%
  transmute(
    min_like = parse_num(klike), min_app = parse_num(kapp),
    mod_like = parse_num(hlike), mod_app = parse_num(happ),
    maj_like = parse_num(xlike), maj_app = parse_num(xapp)
  )

like_items_insp = filtered_by_age %>%
  transmute(
    min_like = parse_num(slike), min_app = parse_num(sapp),
    mod_like = parse_num(flike), mod_app = parse_num(fapp),
    maj_like = parse_num(ylike), maj_app = parse_num(yapp)
  )

psych::alpha(like_items_inv,  check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.7502138 0.7481344
```

```
psych::alpha(like_items_insp, check.keys = TRUE)$total %>% select(raw_alpha, std.alpha) %>% print()
```

```
##  raw_alpha std.alpha
##  0.8304724  0.830882
```

```
#Inter-composite correlation
cor.test(tl_dat$Integrity, tl_dat$likeability, use = "complete.obs") %>% tidy() %>% print()
```

```
## # A tibble: 1 × 8
##   estimate statistic p.value parameter conf.low conf.high method     alternative
##      <dbl>     <dbl>   <dbl>     <int>    <dbl>     <dbl> <chr>      <chr>
## 1    0.803      69.8       0      2676    0.789     0.816 Pearson's… two.sided
```

STEP 3: Model-building strategy

The analyses below use the preregistered step-up structure for each outcome. Models are fit using maximum likelihood (REML = FALSE) so that fixed-effect models can be compared with likelihood-ratio tests. Random slopes are evaluated first for the fully specified interaction model and are stepped down only when the more complex structure is singular. The resulting random-effects structure is then held constant across Models 0–4 so that the likelihood-ratio comparisons test changes in the fixed-effects structure rather than changes in the random-effects structure.

Model sequence for Integrity and Likeability

- Model 0: Null model
- Model 1: Role
- Model 2: QRP Severity
- Model 3: Role + QRP Severity
- Model 4: Role × QRP Severity

Model 1 tests the main effect of researcher role, corresponding to H1a for Integrity and H2a for Likeability. Model 4 tests the Role × QRP Severity interaction, corresponding to H1b and H2b.

```
library(lme4)
library(lmerTest)
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(knitr)
library(kableExtra)

# ------------------------------------------------------------
# Helper functions
# ------------------------------------------------------------

# Extract fixed effects, including Satterthwaite df, t, and p.
get_fixed_effects = function(model) {
  cf = summary(model)$coefficients

  data.frame(
    Parameter = rownames(cf),
    b = cf[, "Estimate"],
    SE = cf[, "Std. Error"],
    df = cf[, "df"],
    t = cf[, "t value"],
    p = cf[, "Pr(>|t|)"],
    row.names = NULL,
    check.names = FALSE
  )
}

# APA-friendly p-value formatting.
fmt_p = function(p) {
  if (is.na(p)) return("")
  if (p < .001) return("< .001")
  formatC(p, digits = 3, format = "f")
}

# APA-friendly number formatting.
fmt_num = function(x, digits = 2) {
  if (length(x) == 0 || is.na(x)) return("")
  formatC(x, digits = digits, format = "f")
}

# Rename model terms for tables.
relabel_terms = function(x) {
  dplyr::recode(
    x,
    "(Intercept)" = "Intercept",
    "role_ec" = "Role",
    "severity_num" = "QRP Severity",
    "role_ec:severity_num" = "Role × QRP Severity",
    .default = x
  )
}

# Likelihood-ratio comparison of two nested ML models.
compare_models = function(previous, current) {
  lr = anova(previous, current)
  data.frame(
    Chisq = lr$Chisq[2],
    df = lr$Df[2],
    p = lr$`Pr(>Chisq)`[2]
  )
}

# Build a model-fit table. Each chi-square is the LR test against
# the immediately preceding model.
model_fit_table = function(models, names) {
  out = lapply(seq_along(models), function(i) {
    m = models[[i]]

    if (i == 1) {
      chisq = NA_real_
      df = NA_real_
      p = NA_real_
    } else {
      lr = compare_models(models[[i - 1]], m)
      chisq = lr$Chisq
      df = lr$df
      p = lr$p
    }

    data.frame(
      Model = names[i],
      AIC = AIC(m),
      BIC = BIC(m),
      logLik = as.numeric(logLik(m)),
      Chisq = chisq,
      df = df,
      p = p,
      check.names = FALSE
    )
  })
```

```
  dplyr::bind_rows(out) %>%
    mutate(
      AIC = round(AIC, 1),
      BIC = round(BIC, 1),
      logLik = round(logLik, 1),
      Chisq = ifelse(is.na(Chisq), "—", sprintf("%.2f", Chisq)),
      df = ifelse(is.na(df), "—", as.character(df)),
      p = ifelse(is.na(p), "—", vapply(p, fmt_p, character(1)))
    )
}

# Extract variance components.
get_variance_components = function(model) {
  vc = as.data.frame(VarCorr(model))
  vc %>%
    select(grp, vcov, sdcor) %>%
    mutate(
      Variance = round(vcov, 3),
      SD = round(sdcor, 3)
    ) %>%
    select(grp, Variance, SD)
}

# ICC for random-intercept components. Random slopes/covariances are not
# interpreted as simple ICC components here.
get_random_intercept_icc = function(model) {
  vc = as.data.frame(VarCorr(model))
  intercepts = vc %>%
    filter(is.na(var1) | var1 == "(Intercept)")

  residual = vc$vcov[vc$grp == "Residual"]
  total = sum(intercepts$vcov) + residual

  intercepts %>%
    mutate(ICC = round(vcov / total, 3)) %>%
    select(grp, ICC)
}
```

STEP 4: Research Question 1 — Integrity

Research Question 1: How is integrity influenced by role and QRP severity?

Hypothesis 1a: Inspectors will be rated higher in integrity than investigators.

Hypothesis 1b: The relative integrity of inspectors versus investigators will increase as QRP severity increases.

Step 4a: Random-effects structure for Integrity

```
# ------------------------------------------------------------
# Maximal participant-level structure
# ------------------------------------------------------------
T4_max1 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
T4_max1_singular = isSingular(T4_max1, tol = 1e-4)
print(T4_max1_singular)
```

```
## [1] TRUE
```

```
# ------------------------------------------------------------
# Step down: remove random interaction slope
# ------------------------------------------------------------
T4_max2 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec + severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
T4_max2_singular = isSingular(T4_max2, tol = 1e-4)
print(T4_max2_singular)
```

```
## [1] TRUE
```

```
# ------------------------------------------------------------
# Step down: retain random role slope only
# ------------------------------------------------------------
T4_max3 = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

T4_max3_singular = isSingular(T4_max3, tol = 1e-4)
print(T4_max3_singular)
```

```
## [1] FALSE
```

```
# ------------------------------------------------------------
# Test preregistered vignette random intercept with the winning
# participant-level structure.
# ------------------------------------------------------------
T4_prereg = lmer(
  Integrity ~ role_ec * severity_num +
    (1 + role_ec | responseid) +
    (1 | vignette_id),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

T4_prereg_singular = isSingular(T4_prereg, tol = 1e-4)
print(T4_prereg_singular)
```

```
## [1] FALSE
```

```
# Decision rule used throughout the analysis:
# retain the vignette intercept if the preregistered structure is
# non-singular; otherwise use the participant-only structure.
Integrity_uses_vignette_term = !T4_prereg_singular

cat(
  "Integrity final random-effects structure includes vignette intercept:",
  Integrity_uses_vignette_term,
  "\n"
)
```

```
## Integrity final random-effects structure includes vignette intercept: TRUE
```

Step 4b: Integrity step-up models

```
# ------------------------------------------------------------
# Fit Models 0–4 using ONE identical random-effects structure.
# ------------------------------------------------------------

if (Integrity_uses_vignette_term) {

  random_part_Integrity = "vignette"

  M0_Integrity = lmer(
    Integrity ~ 1 +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T1 = lmer(
    Integrity ~ role_ec +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T2 = lmer(
    Integrity ~ severity_num +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T3 = lmer(
    Integrity ~ role_ec + severity_num +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T4 = T4_prereg

} else {

  random_part_Integrity = "participant_only"

  M0_Integrity = lmer(
    Integrity ~ 1 +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T1 = lmer(
    Integrity ~ role_ec +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T2 = lmer(
    Integrity ~ severity_num +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T3 = lmer(
    Integrity ~ role_ec + severity_num +
      (1 + role_ec | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  T4 = T4_max3
}
```

```
# Singularity checks for final models.
cat("Integrity final model singularity:\n")
```

```
## Integrity final model singularity:
```

```
print(isSingular(M0_Integrity))
```

```
## [1] FALSE
```

```
print(isSingular(T1))
```

```
## [1] FALSE
```

```
print(isSingular(T2))
```

```
## [1] FALSE
```

```
print(isSingular(T3))
```

```
## [1] FALSE
```

```
print(isSingular(T4))
```

```
## [1] FALSE
```

```
# Complete step-up comparison, including Model 0.
integrity_models = list(
  M0_Integrity,
  T1,
  T2,
  T3,
  T4
)

integrity_model_names = c(
  "Model 0: Null",
  "Model 1: Role",
  "Model 2: QRP Severity",
  "Model 3: Role + QRP Severity",
  "Model 4: Role × QRP Severity"
)

integrity_fit = model_fit_table(
  integrity_models,
  integrity_model_names
)

print(integrity_fit)
```

```
##                          Model    AIC    BIC  logLik Chisq df      p
## 1               Model 0: Null 5826.3 5861.6 -2907.1     —  —      —
## 2               Model 1: Role 5815.5 5856.7 -2900.7 12.80  1 < .001
## 3        Model 2: QRP Severity 5828.1 5869.3 -2907.0  0.00  0      —
## 4 Model 3: Role + QRP Severity 5815.3 5862.5 -2899.7 14.72  1 < .001
## 5 Model 4: Role × QRP Severity 5805.8 5858.9 -2893.9 11.52  1 < .001
```

```
# Full model summaries.
summary(T1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec + (1 + role_ec | responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5815.5    5856.7   -2900.7    5801.5      2671
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9305 -0.5336  0.0050  0.4975  5.4349
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 0.1850   0.4301
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##             Estimate Std. Error     df t value Pr(>|t|)
## (Intercept)   3.1306     0.1762 6.0354  17.764 1.93e-06 ***
## role_ec       1.1764     0.1770 6.1462   6.645 0.000508 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##         (Intr)
## role_ec 0.002
```

```
summary(T2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ severity_num + (1 + role_ec | responseid) + (1 |
##     vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5828.1    5869.3   -2907.0    5814.1      2671
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9298 -0.5329  0.0068  0.4982  5.4377
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 1.5090   1.2284
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1303     0.5017  5.9881   6.239 0.000791 ***
## severity_num  -0.2891     0.6144  5.9837  -0.471 0.654622
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
summary(T3)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec + severity_num + (1 + role_ec | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5815.3    5862.5   -2899.7    5799.3      2670
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9304 -0.5339  0.0084  0.5009  5.4362
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.0410   0.2025
##              role_ec     0.1679   0.4098   0.32
##  vignette_id (Intercept) 0.1292   0.3595
##  Residual                0.3755   0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1306     0.1476  6.0505  21.217 6.54e-07 ***
## role_ec        1.1764     0.1485  6.2091   7.921 0.000181 ***
## severity_num  -0.2891     0.1803  5.9997  -1.603 0.160023
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c
## role_ec     0.003
## severity_nm 0.000  0.000
```

```
summary(T4)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec * severity_num + (1 + role_ec | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5805.8    5858.9   -2893.9    5787.8      2669
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9277 -0.5336  0.0115  0.4970  5.4488
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.04098  0.2024
##              role_ec     0.16783  0.4097   0.32
##  vignette_id (Intercept) 0.01819  0.1349
##  Residual                0.37551  0.6128
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           3.13057    0.05713  6.33858  54.794 1.00e-09 ***
## role_ec               1.17638    0.05956  7.48480  19.750 9.94e-08 ***
## severity_num         -0.28909    0.06899  5.98853  -4.191  0.00577 **
## role_ec:severity_num  0.40805    0.06899  5.98864   5.915  0.00105 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c svrty_
## role_ec     0.017
## severity_nm 0.000  0.000
## rl_c:svrty_ 0.000  0.000  0.000
```

Step 4c: Integrity fixed effects and planned role contrasts

```
# Fixed effects from every model, including Satterthwaite df, t, and p.
integrity_fixed_effects = lapply(
  integrity_models,
  get_fixed_effects
)

names(integrity_fixed_effects) = paste0("Model_", 0:4)

print(integrity_fixed_effects)
```

```
## $Model_0
##     Parameter        b        SE       df        t            p
## 1 (Intercept) 3.130303 0.5109191 5.989118 6.126807 0.0008699531
##
## $Model_1
##     Parameter        b        SE       df         t            p
## 1 (Intercept) 3.130567 0.1762325 6.035446 17.763839 1.933673e-06
## 2     role_ec 1.176382 0.1770355 6.146161  6.644893 5.076029e-04
##
## $Model_2
##      Parameter          b        SE       df          t            p
## 1  (Intercept)  3.1302929 0.5017281 5.988079  6.2390223 0.0007910276
## 2 severity_num -0.2890974 0.6143771 5.983693 -0.4705537 0.6546220311
##
## $Model_3
##      Parameter          b        SE       df         t            p
## 1  (Intercept)  3.1305677 0.1475504 6.050460 21.216941 6.544313e-07
## 2      role_ec  1.1763829 0.1485084 6.209082  7.921321 1.809007e-04
## 3 severity_num -0.2890992 0.1803309 5.999654 -1.603159 1.600230e-01
##
## $Model_4
##              Parameter          b         SE       df         t            p
## 1          (Intercept)  3.1305668 0.05713328 6.338584 54.794106 1.004539e-09
## 2              role_ec  1.1763834 0.05956286 7.484802 19.750285 9.935063e-08
## 3         severity_num -0.2890927 0.06898510 5.988534 -4.190654 5.769390e-03
## 4 role_ec:severity_num  0.4080529 0.06898539 5.988637  5.915062 1.046426e-03
```

```
# Planned comparison: inspectors versus investigators at each severity.
# Bonferroni correction is applied across the three severity-specific
# role comparisons.
emm_Integrity = emmeans(
  T4,
  ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1))
)

emm_Integrity_role = emmeans(
  T4,
  pairwise ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_Integrity_role$contrasts)
```

```
## severity_num = -1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -1.54 0.308 16.7  -4.997  0.0001
##
## severity_num =  0:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -2.35 0.197 16.1 -11.956 <0.0001
##
## severity_num =  1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1    -3.17 0.307 16.7 -10.305 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

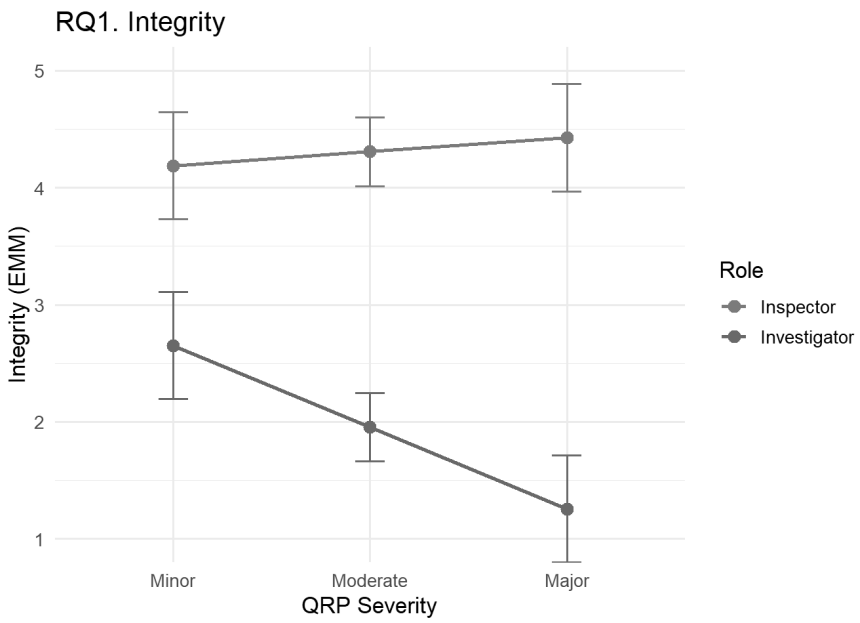
```
# Cohen's d for each role contrast.
eff_size(
  emm_Integrity,
  sigma = sigma(T4),
  edf = df.residual(T4)
)
```

```
## severity_num = -1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -2.51 0.503 16.8    -3.57    -1.45
##
## severity_num =  0:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -3.84 0.325 16.3    -4.53    -3.15
##
## severity_num =  1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -5.17 0.507 16.8    -6.24    -4.10
##
## sigma used for effect sizes: 0.6128
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

Step 4d: Integrity figure

```
emm_Integrity_df = as.data.frame(
  emmeans(
    T4,
    ~ role_ec * severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Role = if_else(
      role_ec == 1,
      "Inspector",
      "Investigator"
    ),
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_Integrity_df,
  aes(
    x = Severity,
    y = emmean,
    color = Role,
    group = Role
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.15
  ) +
  scale_color_manual(
    values = c(
      "Inspector" = "#2196F3",
      "Investigator" = "#E53935"
    )
  ) +
  labs(
    title = "RQ1. Integrity",
    y = "Integrity (EMM)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 13) +
  coord_cartesian(ylim = c(1, 5))
```



STEP 5: Research Question 2 — Likeability

Research Question 2: How is likeability influenced by role and QRP severity?

Hypothesis 2a: Investigators will be rated more likeable than inspectors.

Hypothesis 2b: At low QRP severity, investigators will be rated more likeable than inspectors, but at high QRP severity, inspectors will be rated more likeable than investigators.

Step 5a: Random-effects structure for Likeability

```
# ------------------------------------------------------------
# Maximal participant-level structure
# ------------------------------------------------------------
L4_max1 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_max1_singular = isSingular(L4_max1, tol = 1e-4)
print(L4_max1_singular)
```

```
## [1] FALSE
```

```
# Step down: remove random interaction slope.
L4_max2 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec + severity_num | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
L4_max2_singular = isSingular(L4_max2, tol = 1e-4)
print(L4_max2_singular)
```

```
## [1] TRUE
```

```
# Step down: retain random role slope only.
L4_max3 = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec | responseid),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_max3_singular = isSingular(L4_max3, tol = 1e-4)
print(L4_max3_singular)
```

```
## [1] FALSE
```

```
# Test preregistered vignette random intercept with the maximal
# participant-level structure.
L4_prereg = lmer(
  likeability ~ role_ec * severity_num +
    (1 + role_ec * severity_num | responseid) +
    (1 | vignette_id),
  data = tl_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

L4_prereg_singular = isSingular(L4_prereg, tol = 1e-4)
print(L4_prereg_singular)
```

```
## [1] FALSE
```

```
likeability_uses_vignette_term = !L4_prereg_singular

cat(
  "Likeability final random-effects structure includes vignette intercept:",
  likeability_uses_vignette_term,
  "\n"
)
```

```
## Likeability final random-effects structure includes vignette intercept: TRUE
```

Step 5b: Likeability step-up models

```
# ------------------------------------------------------------
# Fit Models 0–4 using ONE identical random-effects structure.
# ------------------------------------------------------------

if (likeability_uses_vignette_term) {

  random_part_likeability = "participant_maximal_plus_vignette"

  M0_like = lmer(
    likeability ~ 1 +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L1 = lmer(
    likeability ~ role_ec +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L2 = lmer(
    likeability ~ severity_num +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L3 = lmer(
    likeability ~ role_ec + severity_num +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L4 = L4_prereg

} else {

  random_part_likeability = "participant_maximal"

  M0_like = lmer(
    likeability ~ 1 +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L1 = lmer(
    likeability ~ role_ec +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L2 = lmer(
    likeability ~ severity_num +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

  L3 = lmer(
    likeability ~ role_ec + severity_num +
      (1 + role_ec * severity_num | responseid),
    data = tl_dat,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
```

```
  L4 = L4_max1
}
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -8.1e+00
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -1.7e+01
```

```
cat("Likeability final model singularity:\n")
```

```
## Likeability final model singularity:
```

```
print(isSingular(M0_like))
```

```
## [1] TRUE
```

```
print(isSingular(L1))
```

```
## [1] FALSE
```

```
print(isSingular(L2))
```

```
## [1] FALSE
```

```
print(isSingular(L3))
```

```
## [1] TRUE
```

```
print(isSingular(L4))
```

```
## [1] FALSE
```

```
likeability_models = list(
  M0_like,
  L1,
  L2,
  L3,
  L4
)

likeability_model_names = c(
  "Model 0: Null",
  "Model 1: Role",
  "Model 2: QRP Severity",
  "Model 3: Role + QRP Severity",
  "Model 4: Role × QRP Severity"
)

likeability_fit = model_fit_table(
  likeability_models,
  likeability_model_names
)

print(likeability_fit)
```

```
##                         Model    AIC    BIC  logLik Chisq df      p
## 1               Model 0: Null 6191.7 6268.3 -3082.8     —  —      —
## 2               Model 1: Role 6183.5 6266.0 -3077.7 10.20  1  0.001
## 3       Model 2: QRP Severity 6193.5 6276.0 -3082.7  0.00  0      —
## 4 Model 3: Role + QRP Severity 6184.5 6272.9 -3077.2 10.99  1 < .001
## 5 Model 4: Role × QRP Severity 6169.5 6263.8 -3068.8 16.97  1 < .001
```

```
summary(L1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec + (1 + role_ec * severity_num | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6183.5    6266.0   -3077.7    6155.5      2664
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9403 -0.4843  0.0081  0.5067  4.1002
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.07706  0.27760
##              role_ec              0.24482  0.49480   0.26
##              severity_num         0.00554  0.07443  -0.53 -0.94
##              role_ec:severity_num 0.04783  0.21869  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.09784  0.31280
##  Residual                         0.35920  0.59933
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##             Estimate Std. Error     df t value Pr(>|t|)
## (Intercept)   3.1358     0.1289 6.0788  24.328 2.73e-07 ***
## role_ec       0.6663     0.1303 6.3558   5.112  0.00185 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##         (Intr)
## role_ec 0.005
```

```
summary(L2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## likeability ~ severity_num + (1 + role_ec * severity_num | responseid) +
##     (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6193.5    6276.0   -3082.7    6165.5      2664
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9395 -0.4840  0.0108  0.5061  4.1041
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.07707  0.27761
##              role_ec              0.24487  0.49484   0.26
##              severity_num         0.00554  0.07443  -0.53 -0.94
##              role_ec:severity_num 0.04781  0.21865  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.52003  0.72113
##  Residual                         0.35920  0.59934
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1353     0.2949  5.9543  10.631 4.29e-05 ***
## severity_num  -0.1642     0.3609  5.9303  -0.455    0.665
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
summary(L3)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec + severity_num + (1 + role_ec * severity_num |
##     responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6184.5    6272.9   -3077.2    6154.5      2663
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8763 -0.4812  0.0096  0.5046  4.1504
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.077037 0.27756
##              role_ec              0.244774 0.49475   0.26
##              severity_num         0.005438 0.07374  -0.53 -0.96
##              role_ec:severity_num 0.047782 0.21859  -0.02  0.23 -0.19
##  vignette_id (Intercept)          0.079666 0.28225
##  Residual                         0.359370 0.59947
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.1359     0.1166  6.0920  26.906 1.45e-07 ***
## role_ec        0.6666     0.1181  6.4325   5.642  0.00105 **
## severity_num  -0.1647     0.1419  5.9457  -1.161  0.29016
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c
## role_ec      0.006
## severity_nm -0.001 -0.005
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
summary(L4)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: likeability ~ role_ec * severity_num + (1 + role_ec * severity_num |
##     responseid) + (1 | vignette_id)
##    Data: tl_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    6169.5    6263.8   -3068.8    6137.5      2662
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9512 -0.4872  0.0110  0.5102  4.1226
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.076949 0.27740
##              role_ec              0.244494 0.49446   0.26
##              severity_num         0.005531 0.07437  -0.53 -0.94
##              role_ec:severity_num 0.047707 0.21842  -0.03  0.23 -0.33
##  vignette_id (Intercept)          0.004040 0.06356
##  Residual                         0.359310 0.59942
## Number of obs: 2678, groups:  responseid, 448; vignette_id, 6
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           3.13589    0.03130  8.50866 100.189 2.26e-14 ***
## role_ec               0.66755    0.03680 16.13070  18.140 3.72e-12 ***
## severity_num         -0.16481    0.03498  5.91609  -4.711  0.00341 **
## role_ec:severity_num  0.33805    0.03631  6.85648   9.310 3.89e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) role_c svrty_
## role_ec      0.070
## severity_nm -0.023 -0.060
## rl_c:svrty_ -0.003  0.041 -0.009
```

Step 5c: Likeability fixed effects and planned role contrasts

```
likeability_fixed_effects = lapply(
  likeability_models,
  get_fixed_effects
)

names(likeability_fixed_effects) = paste0("Model_", 0:4)

print(likeability_fixed_effects)
```

```
## $Model_0
##     Parameter        b        SE       df        t            p
## 1 (Intercept) 3.135266 0.2999751 5.955134 10.45176 4.722539e-05
##
## $Model_1
##     Parameter         b        SE       df         t            p
## 1 (Intercept) 3.1357621 0.1288942 6.078846 24.328189 2.734277e-07
## 2     role_ec 0.6662713 0.1303398 6.355754  5.111804 1.849241e-03
##
## $Model_2
##      Parameter          b        SE       df          t            p
## 1  (Intercept)  3.1352764 0.2949213 5.954267 10.6308930 4.291348e-05
## 2 severity_num -0.1641912 0.3608626 5.930343 -0.4549963 6.652929e-01
##
## $Model_3
##      Parameter          b        SE       df         t            p
## 1  (Intercept)  3.1359306 0.1165518 6.091968 26.905897 1.451691e-07
## 2      role_ec  0.6666303 0.1181490 6.432463  5.642283 1.047393e-03
## 3 severity_num -0.1647117 0.1418814 5.945726 -1.160911 2.901565e-01
##
## $Model_4
##              Parameter          b         SE        df          t            p
## 1          (Intercept)  3.1358873 0.03129985  8.508664 100.188566 2.262092e-14
## 2              role_ec  0.6675478 0.03680049 16.130700  18.139642 3.720133e-12
## 3         severity_num -0.1648123 0.03498387  5.916089  -4.711094 3.412601e-03
## 4 role_ec:severity_num  0.3380490 0.03630913  6.856480   9.310302 3.891875e-05
```

```
# Planned role comparisons at each severity.
emm_like_role = emmeans(
  L4,
  pairwise ~ role_ec | severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_like_role$contrasts)
```

```
## severity_num = -1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -0.659 0.154 13.3  -4.275  0.0009
##
## severity_num =  0:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -1.335 0.104 12.6 -12.834 <0.0001
##
## severity_num =  1:
##  contrast               estimate    SE   df t.ratio p.value
##  (role_ec-1) - role_ec1   -2.011 0.157 13.0 -12.814 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

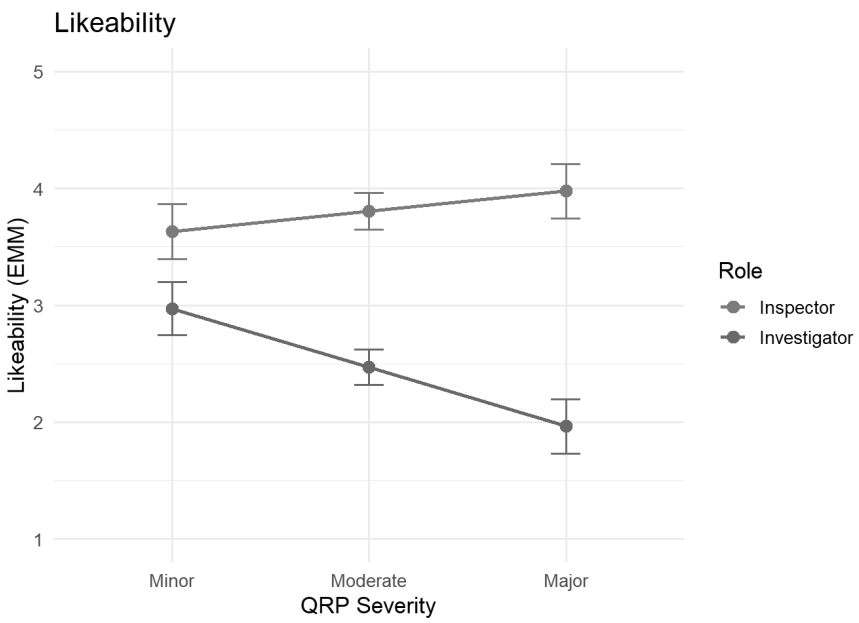
```
# Cohen's d for each role comparison.
eff_size(
  emmeans(
    L4,
    ~ role_ec | severity_num,
    at = list(severity_num = c(-1, 0, 1))
  ),
  sigma = sigma(L4),
  edf = df.residual(L4)
)
```

```
## severity_num = -1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -1.10 0.258 13.3    -1.65   -0.544
##
## severity_num =  0:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -2.23 0.176 12.7    -2.61   -1.846
##
## severity_num =  1:
##  contrast               effect.size    SE   df lower.CL upper.CL
##  (role_ec-1) - role_ec1       -3.36 0.266 13.3    -3.93   -2.782
##
## sigma used for effect sizes: 0.5994
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

Step 5d: Likeability figure

```
emm_like_df = as.data.frame(
  emmeans(
    L4,
    ~ role_ec * severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Role = if_else(
      role_ec == 1,
      "Inspector",
      "Investigator"
    ),
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_like_df,
  aes(
    x = Severity,
    y = emmean,
    color = Role,
    group = Role
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.15
  ) +
  scale_color_manual(
    values = c(
      "Inspector" = "#2196F3",
      "Investigator" = "#E53935"
    )
  ) +
  labs(
    title = "Likeability",
    y = "Likeability (EMM)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 13) +
  coord_cartesian(ylim = c(1, 5))
```



STEP 6: Research Question 3 — Perceived Trueness

Research Question 3: Is the perceived trueness of the finding influenced by QRP severity?

Hypothesis 3: Perceived trueness will decrease as QRP severity increases.

Step 6a: Trueness data and model

```
# Trueness is only rated for investigators, with one item per vignette.
trueness_dat = filtered_by_age %>%
  transmute(
    responseid,
    minor = parse_num(kcon),
    moderate = parse_num(hcon),
    major = parse_num(xcon)
  ) %>%
  pivot_longer(
    cols = c(minor, moderate, major),
    names_to = "severity",
    values_to = "trueness"
  ) %>%
  mutate(
    severity = factor(
      severity,
      levels = c("minor", "moderate", "major")
    ),
    severity_num = case_when(
      severity == "minor" ~ -1,
      severity == "moderate" ~ 0,
      severity == "major" ~ 1
    ),
    responseid = factor(responseid),
    trueness_c = trueness - 3
  )


# Descriptives.
trueness_descriptives = trueness_dat %>%
  group_by(severity) %>%
  summarise(
    n = sum(!is.na(trueness)),
    M = round(mean(trueness, na.rm = TRUE), 2),
    SD = round(sd(trueness, na.rm = TRUE), 2),
    .groups = "drop"
  )

print(trueness_descriptives)
```

```
## # A tibble: 3 × 4
##   severity     n     M    SD
##   <fct>    <int> <dbl> <dbl>
## 1 minor      446  2.98  0.91
## 2 moderate   447  1.86  0.92
## 3 major      447  1.53  0.88
```

```
# FT1: perceived trueness predicted by QRP severity.
FT1 = lmer(
  trueness_c ~ severity_num +
    (1 | responseid) +
    (1 | severity),
  data = trueness_dat,
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

summary(FT1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: trueness_c ~ severity_num + (1 | responseid) + (1 | severity)
##    Data: trueness_dat
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    3503.7    3529.7   -1746.8    3493.7      1335
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7309 -0.6556 -0.0754  0.4459  4.0079
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  responseid (Intercept) 0.16689  0.4085
##  severity   (Intercept) 0.03313  0.1820
##  Residual               0.65229  0.8076
## Number of obs: 1340, groups:  responseid, 448; severity, 3
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.8771     0.1091  3.1960  -8.040  0.00318 **
## severity_num  -0.7213     0.1315  2.9993  -5.485  0.01193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## severity_nm 0.000
```

```
isSingular(FT1)
```

```
## [1] FALSE
```

```
# Fixed effects, including Satterthwaite df, t, and p.
trueness_fixed_effects = get_fixed_effects(FT1)
print(trueness_fixed_effects)
```

```
##      Parameter          b        SE       df         t           p
## 1  (Intercept) -0.8770718 0.1090937 3.195991 -8.039618 0.003179199
## 2 severity_num -0.7212843 0.1315047 2.999292 -5.484857 0.011928040
```

```
# Planned comparisons across severity levels.
emm_trueness = emmeans(
  FT1,
  pairwise ~ severity_num,
  at = list(severity_num = c(-1, 0, 1)),
  adjust = "bonferroni"
)

summary(emm_trueness$contrasts)
```

```
##  contrast                        estimate    SE   df t.ratio p.value
##  (severity_num-1) - severity_num0    0.721 0.227 8.81   3.183  0.0343
##  (severity_num-1) - severity_num1    1.443 0.453 8.81   3.183  0.0343
##  severity_num0 - severity_num1       0.721 0.227 8.81   3.183  0.0343
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

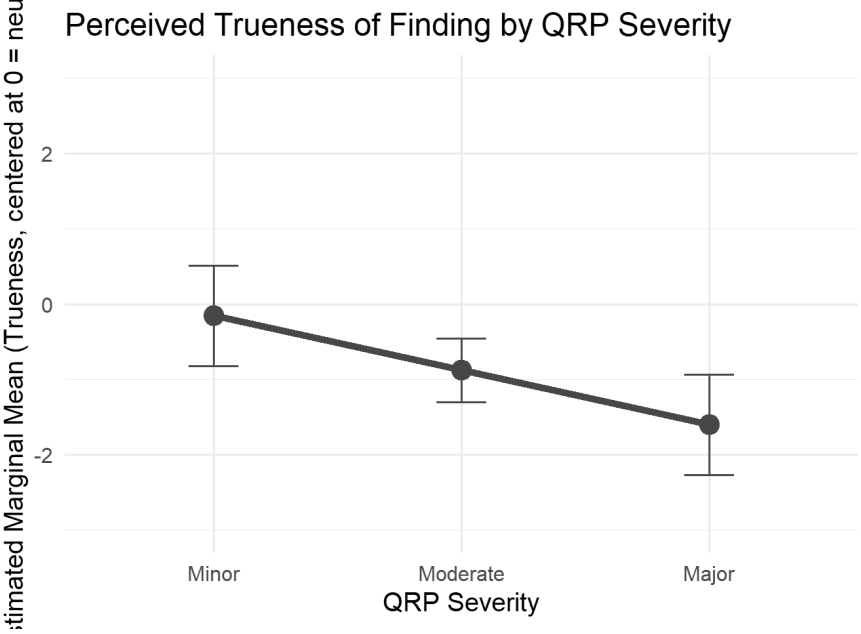
```
eff_size(
  emmeans(
    FT1,
    ~ severity_num,
    at = list(severity_num = c(-1, 0, 1))
  ),
  sigma = sigma(FT1),
  edf = df.residual(FT1)
)
```

```
##  contrast                        effect.size    SE   df lower.CL upper.CL
##  (severity_num-1) - severity_num0       0.893 0.281 8.64    0.253     1.53
##  (severity_num-1) - severity_num1       1.786 0.562 8.74    0.509     3.06
##  severity_num0 - severity_num1          0.893 0.281 8.64    0.253     1.53
##
## sigma used for effect sizes: 0.8076
## Degrees-of-freedom method: inherited from kenward-roger when re-gridding
## Confidence level used: 0.95
```

Step 6b: Trueness figure

```
emm_true_df = as.data.frame(
  emmeans(
    FT1,
    ~ severity_num,
    at = list(severity_num = c(-1, 0, 1))
  )
) %>%
  mutate(
    Severity = factor(
      severity_num,
      levels = c(-1, 0, 1),
      labels = c("Minor", "Moderate", "Major")
    )
  )

ggplot(
  emm_true_df,
  aes(
    x = Severity,
    y = emmean,
    group = 1
  )
) +
  geom_line(linewidth = 2, color = "#7B1FA2") +
  geom_point(size = 5, color = "#7B1FA2") +
  geom_errorbar(
    aes(ymin = lower.CL, ymax = upper.CL),
    width = 0.20,
    color = "#7B1FA2"
  ) +
  labs(
    title = "Perceived Trueness of Finding by QRP Severity",
    y = "Estimated Marginal Mean (Trueness, centered at 0 = neutral)",
    x = "QRP Severity"
  ) +
  theme_minimal(base_size = 15) +
  coord_cartesian(ylim = c(-3, 3))
```


Perceived Trueness of Finding by QRP Severity
stimated Marginal Mean (Trueness, centered at 0 = neu
2
0
-2
Minor
Moderate
Major
QRP Severity

STEP 7: Robustness checks — demographic covariates

```
# Attach demographics to tl_dat.
demo_dat = filtered_by_age %>%
  transmute(
    responseid = responseid,
    gender = factor(`gender.identity`),
    political = factor(
      political,
      levels = c(
        "Strongly Liberal",
        "Liberal",
        "Lean Liberal",
        "Centrist",
        "Lean Conservative",
        "Conservative",
        "Strongly Conservative"
      )
    ),
    religion = factor(religion)
  )

tl_robust = tl_dat %>%
  mutate(responseid = as.character(responseid)) %>%
  left_join(demo_dat, by = "responseid") %>%
  mutate(responseid = factor(responseid))


# Integrity robustness model uses the exact random-effects structure
# selected for the final Integrity Model 4.
if (Integrity_uses_vignette_term) {

  T4_robust = lmer(
    Integrity ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec | responseid) +
      (1 | vignette_id),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

} else {

  T4_robust = lmer(
    Integrity ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec | responseid),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
}


# Likeability robustness model uses the exact random-effects structure
# selected for the final Likeability Model 4.
if (likeability_uses_vignette_term) {

  L4_robust = lmer(
    likeability ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec * severity_num | responseid) +
      (1 | vignette_id),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )

} else {

  L4_robust = lmer(
    likeability ~ role_ec * severity_num +
      gender + political + religion +
      (1 + role_ec * severity_num | responseid),
    data = tl_robust,
    REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
  )
}
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: Model failed to converge with 1 negative eigenvalue: -3.2e-01
```

```
summary(T4_robust)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Integrity ~ role_ec * severity_num + gender + political + religion +
##     (1 + role_ec | responseid) + (1 | vignette_id)
##    Data: tl_robust
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5599.2    5733.8   -2776.6    5553.2      2551
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9084 -0.5233  0.0089  0.4967  5.4390
##
## Random effects:
##  Groups      Name        Variance Std.Dev. Corr
##  responseid  (Intercept) 0.03990  0.1998
##              role_ec     0.16828  0.4102   0.33
##  vignette_id (Intercept) 0.01821  0.1350
##  Residual                0.37452  0.6120
## Number of obs: 2574, groups:  responseid, 429; vignette_id, 6
##
## Fixed effects:
##                                  Estimate Std. Error         df t value
## (Intercept)                       3.153051   0.101994  59.533220  30.914
## role_ec                           1.177933   0.059775   7.550335  19.706
## severity_num                     -0.292735   0.069073   5.986598  -4.238
## genderNon-binary / third gender  -0.277020   0.230884 428.135496  -1.200
## genderPrefer not to say          -0.042871   0.231147 428.135496  -0.185
## genderWoman                      -0.015940   0.040382 428.135496  -0.395
## politicalLiberal                 -0.046542   0.082963 428.135496  -0.561
## politicalLean Liberal            -0.099518   0.088391 428.135496  -1.126
## politicalCentrist                -0.072534   0.079578 428.135497  -0.911
## politicalLean Conservative       -0.071195   0.082430 428.135497  -0.864
## politicalConservative            -0.024063   0.078867 428.135497  -0.305
## politicalStrongly Conservative   -0.006314   0.086966 428.135497  -0.073
## religionIslam                     0.309681   0.319419 428.135495   0.970
## religionJudaism                   0.046641   0.226141 428.135495   0.206
## religionNot Affiliated           -0.009012   0.057244 428.135496  -0.157
## religionOther                     0.135342   0.071001 428.135495   1.906
## religionProtestantism (Christian)   0.060335   0.039364 428.135496   1.533
## role_ec:severity_num              0.416278   0.069073   5.986598   6.027
##                                 Pr(>|t|)
## (Intercept)                      < 2e-16 ***
## role_ec                         9.12e-08 ***
## severity_num                     0.00548 **
## genderNon-binary / third gender  0.23087
## genderPrefer not to say          0.85295
## genderWoman                      0.69325
## politicalLiberal                 0.57510
## politicalLean Liberal            0.26085
## politicalCentrist                0.36256
## politicalLean Conservative       0.38824
## politicalConservative            0.76043
## politicalStrongly Conservative   0.94216
## religionIslam                    0.33284
## religionJudaism                  0.83669
## religionNot Affiliated           0.87498
## religionOther                    0.05729 .
## religionProtestantism (Christian)  0.12608
## role_ec:severity_num             0.00095 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 18 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
summary(L4_robust)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## likeability ~ role_ec * severity_num + gender + political + religion +
##     (1 + role_ec * severity_num | responseid) + (1 | vignette_id)
##    Data: tl_robust
## Control: lmerControl(optimizer = "bobyqa")
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5940.8    6116.4   -2940.4    5880.8      2544
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.7935 -0.4820  0.0202  0.5210  3.7346
##
## Random effects:
##  Groups      Name                 Variance Std.Dev. Corr
##  responseid  (Intercept)          0.071454 0.26731
##              role_ec              0.242165 0.49210   0.26
##              severity_num         0.005723 0.07565  -0.52 -0.96
##              role_ec:severity_num 0.045720 0.21382  -0.02  0.24 -0.21
##  vignette_id (Intercept)          0.003778 0.06147
##  Residual                         0.361073 0.60089
## Number of obs: 2574, groups:  responseid, 429; vignette_id, 6
##
## Fixed effects:
##                                  Estimate Std. Error        df t value
## (Intercept)                       3.04077    0.10041 325.78946  30.283
## role_ec                           0.66608    0.03653  17.13213  18.234
## severity_num                     -0.16725    0.03418   5.91030  -4.893
## genderNon-binary / third gender  -0.04049    0.26151 429.00555  -0.155
## genderPrefer not to say           0.29969    0.26181 429.00555   1.145
## genderWoman                       0.03465    0.04574 429.00556   0.757
## politicalLiberal                  0.05195    0.09397 429.00556   0.553
## politicalLean Liberal            -0.02167    0.10012 429.00556  -0.216
## politicalCentrist                -0.03189    0.09014 429.00556  -0.354
## politicalLean Conservative       -0.01950    0.09337 429.00556  -0.209
## politicalConservative             0.02450    0.08933 429.00556   0.274
## politicalStrongly Conservative    0.06482    0.09850 429.00556   0.658
## religionIslam                     0.51745    0.36179 429.00555   1.430
## religionJudaism                   0.14722    0.25614 429.00555   0.575
## religionNot Affiliated           -0.03877    0.06484 429.00556  -0.598
## religionOther                     0.17098    0.08042 429.00555   2.126
## religionProtestantism (Christian)   0.08186    0.04459 429.00556   1.836
## role_ec:severity_num              0.34557    0.03552   6.88122   9.730
##                                 Pr(>|t|)
## (Intercept)                      < 2e-16 ***
## role_ec                         1.17e-12 ***
## severity_num                     0.00285 **
## genderNon-binary / third gender  0.87702
## genderPrefer not to say          0.25297
## genderWoman                      0.44917
## politicalLiberal                 0.58069
## politicalLean Liberal            0.82875
## politicalCentrist                0.72369
## politicalLean Conservative       0.83470
## politicalConservative            0.78400
## politicalStrongly Conservative   0.51084
## religionIslam                    0.15338
## religionJudaism                  0.56575
## religionNot Affiliated           0.55019
## religionOther                    0.03407 *
## religionProtestantism (Christian)  0.06704 .
## role_ec:severity_num             2.86e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 18 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# Compare interaction coefficients with and without demographic covariates.
cat("Integrity interaction, original:",
    fixef(T4)[["role_ec:severity_num"]], "\n")
```

```
## Integrity interaction, original: 0.4080529
```

```
cat("Integrity interaction, adjusted:",
    fixef(T4_robust)[["role_ec:severity_num"]], "\n")
```

```
## Integrity interaction, adjusted: 0.4162782
```

```
cat("Likeability interaction, original:",
    fixef(L4)[["role_ec:severity_num"]], "\n")
```

```
## Likeability interaction, original: 0.338049
```

```
cat("Likeability interaction, adjusted:",
    fixef(L4_robust)[["role_ec:severity_num"]], "\n")
```

```
## Likeability interaction, adjusted: 0.3455711
```

==========

APA TABLES

==========

```r
# ──────────────────────────────────────────────────────────────
# APA STEP-UP MODEL TABLES
# Side-by-side format
# ──────────────────────────────────────────────────────────────

get_fe = function(model) {

  cf = summary(model)$coefficients

  list(
    b  = setNames(
      round(cf[, "Estimate"], 2),
      rownames(cf)
    ),

    se = setNames(
      round(cf[, "Std. Error"], 2),
      rownames(cf)
    ),

    df = if ("df" %in% colnames(cf)) {
      setNames(
        round(cf[, "df"], 2),
        rownames(cf)
      )
    } else {
      NULL
    },

    t = if ("t value" %in% colnames(cf)) {
      setNames(
        round(cf[, "t value"], 2),
        rownames(cf)
      )
    } else {
      NULL
    },

    p = setNames(
      cf[, "Pr(>|t|)"],
      rownames(cf)
    )
  )
}


# ──────────────────────────────────────────────────────────────
# Format p values
# ──────────────────────────────────────────────────────────────

fmt_p = function(p) {

  if (is.na(p)) return("")

  if (p < .001) {
    return("< .001***")
  }

  stars =
    ifelse(
      p < .01, "**",
      ifelse(
        p < .05, "*",
        ""
      )
    )

  paste0(
    formatC(
      p,
      digits = 3,
      format = "f"
    ),
    stars
  )
}


# ──────────────────────────────────────────────────────────────
# Extract one fixed-effect cell
# ──────────────────────────────────────────────────────────────

cell = function(fe, term) {

  if (!term %in% names(fe$b)) {
    return(
```

```
      c("", "", "")
    )
  }

  c(
    as.character(fe$b[term]),
    as.character(fe$se[term]),
    fmt_p(fe$p[term])
  )
}


# ──────────────────────────────────────────────────────────
# Random effects
# ──────────────────────────────────────────────────────────

get_re = function(model) {

  vc = as.data.frame(
    lme4::VarCorr(model)
  )

  setNames(
    round(vc$vcov, 3),
    vc$grp
  )
}


# ──────────────────────────────────────────────────────────
# ICC
# ──────────────────────────────────────────────────────────

get_icc = function(model) {

  vc = as.data.frame(
    lme4::VarCorr(model)
  )

  total = sum(vc$vcov)

  setNames(
    round(
      vc$vcov / total,
      3
    ),
    vc$grp
  )
}


# ──────────────────────────────────────────────────────────
# STEP-UP TABLE FUNCTION
# ──────────────────────────────────────────────────────────

make_stepup_table = function(
    m0,
    m1,
    m2,
    m3,
    m4,
    caption_text,
    footnote_text
) {

  models = list(
    m0,
    m1,
    m2,
    m3,
    m4
  )

  fe_list = lapply(
    models,
    get_fe
  )

  re_list = lapply(
    models,
    get_re
  )

  icc_list = lapply(
    models,
    get_icc
```

```
)


# ──────────────────────────────────────────────────────────
# Fixed-effect terms
# ──────────────────────────────────────────────────────────

term_labels = c(
  "(Intercept)" =
    "Intercept",

  "role_ec" =
    "Role",

  "severity_num" =
    "QRP Severity",

  "role_ec:severity_num" =
    "Role × QRP Severity"
)


rows = lapply(
  names(term_labels),
  function(term) {

    cells =
      do.call(
        cbind,
        lapply(
          fe_list,
          function(fe) cell(fe, term)
        )
      )

    data.frame(

      Parameter =
        term_labels[term],

      M0_b =
        cells[1, 1],

      M0_se =
        cells[2, 1],

      M0_p =
        cells[3, 1],

      M1_b =
        cells[1, 2],

      M1_se =
        cells[2, 2],

      M1_p =
        cells[3, 2],

      M2_b =
        cells[1, 3],

      M2_se =
        cells[2, 3],

      M2_p =
        cells[3, 3],

      M3_b =
        cells[1, 4],

      M3_se =
        cells[2, 4],

      M3_p =
        cells[3, 4],

      M4_b =
        cells[1, 5],

      M4_se =
        cells[2, 5],

      M4_p =
        cells[3, 5],

      stringsAsFactors = FALSE
```

```
    )
  }
)


fe_df =
  do.call(
    rbind,
    rows
  )


# ─────────────────────────────────────────────────────────
# Random effects
#
# IMPORTANT:
# Your data use vignette_id, NOT severity.
# ─────────────────────────────────────────────────────────

re_grps = c(

  "responseid" =
    "Participant variance",

  "vignette_id" =
    "Vignette variance",

  "Residual" =
    "Residual variance"
)


re_rows =
  lapply(
    names(re_grps),
    function(grp) {

      vals =
        sapply(
          re_list,
          function(re) {

            if (grp %in% names(re)) {

              as.character(
                re[grp]
              )

            } else {

              "–"
            }
          }
        )

      data.frame(

        Parameter =
          re_grps[grp],

        M0_b = vals[1],
        M0_se = "",
        M0_p = "",

        M1_b = vals[2],
        M1_se = "",
        M1_p = "",

        M2_b = vals[3],
        M2_se = "",
        M2_p = "",

        M3_b = vals[4],
        M3_se = "",
        M3_p = "",

        M4_b = vals[5],
        M4_se = "",
        M4_p = "",

        stringsAsFactors = FALSE
      )
    }
  )
```

```
re_df =
  do.call(
    rbind,
    re_rows
  )


# ──────────────────────────────────────────────────────────
# ICC
# ──────────────────────────────────────────────────────────

icc_rows =
  lapply(
    c(
      "responseid",
      "vignette_id"
    ),
    function(grp) {

      vals =
        sapply(
          icc_list,
          function(ic) {

            if (grp %in% names(ic)) {

              as.character(
                ic[grp]
              )

            } else {

              "–"
            }
          }
        )

      label =
        ifelse(
          grp == "responseid",
          "ICC: Participant",
          "ICC: Vignette"
        )

      data.frame(

        Parameter = label,

        M0_b = vals[1],
        M0_se = "",
        M0_p = "",

        M1_b = vals[2],
        M1_se = "",
        M1_p = "",

        M2_b = vals[3],
        M2_se = "",
        M2_p = "",

        M3_b = vals[4],
        M3_se = "",
        M3_p = "",

        M4_b = vals[5],
        M4_se = "",
        M4_p = "",

        stringsAsFactors = FALSE
      )
    }
  )


icc_df =
  do.call(
    rbind,
    icc_rows
  )


# ──────────────────────────────────────────────────────────
# AIC and BIC
# ──────────────────────────────────────────────────────────

fit_vals =
```

```
  sapply(
    models,
    function(m) {

      c(
        AIC = round(
          AIC(m),
          1
        ),

        BIC = round(
          BIC(m),
          1
        )
      )
    }
  )


fit_df =
  data.frame(

    Parameter =
      c(
        "AIC",
        "BIC"
      ),

    M0_b =
      as.character(
        fit_vals[, 1]
      ),

    M0_se = "",
    M0_p = "",

    M1_b =
      as.character(
        fit_vals[, 2]
      ),

    M1_se = "",
    M1_p = "",

    M2_b =
      as.character(
        fit_vals[, 3]
      ),

    M2_se = "",
    M2_p = "",

    M3_b =
      as.character(
        fit_vals[, 4]
      ),

    M3_se = "",
    M3_p = "",

    M4_b =
      as.character(
        fit_vals[, 5]
      ),

    M4_se = "",
    M4_p = "",

    stringsAsFactors = FALSE
  )


# ──────────────────────────────────────────────────────────
# Model comparison χ²
# ──────────────────────────────────────────────────────────

comparisons = list()

for (i in 2:5) {

  comp =
    anova(
      models[[i - 1]],
      models[[i]]
    )
```

```
  chi =
    comp$Chisq[2]

  df =
    comp$Df[2]

  p =
    comp$`Pr(>Chisq)`[2]

  comparisons[[i - 1]] =
    c(
      chi,
      df,
      p
    )
}


comparison_df =
  data.frame(

    Parameter =
      c(
        "Model comparison χ²",
        "Model comparison df",
        "Model comparison p"
      ),

    M0_b = "—",
    M0_se = "",
    M0_p = "",

    M1_b =
      as.character(
        round(
          comparisons[[1]][1],
          2
        )
      ),

    M1_se =
      as.character(
        comparisons[[1]][2]
      ),

    M1_p =
      fmt_p(
        comparisons[[1]][3]
      ),

    M2_b =
      as.character(
        round(
          comparisons[[2]][1],
          2
        )
      ),

    M2_se =
      as.character(
        comparisons[[2]][2]
      ),

    M2_p =
      fmt_p(
        comparisons[[2]][3]
      ),

    M3_b =
      as.character(
        round(
          comparisons[[3]][1],
          2
        )
      ),

    M3_se =
      as.character(
        comparisons[[3]][2]
      ),

    M3_p =
      fmt_p(
        comparisons[[3]][3]
      ),
```

```
      M4_b =
        as.character(
          round(
            comparisons[[4]][1],
            2
          )
        ),

      M4_se =
        as.character(
          comparisons[[4]][2]
        ),

      M4_p =
        fmt_p(
          comparisons[[4]][3]
        ),

      stringsAsFactors = FALSE
    )


  # ──────────────────────────────────────────────────────────
  # Combine everything
  # ──────────────────────────────────────────────────────────

  full_df =
    rbind(
      fe_df,
      re_df,
      icc_df,
      fit_df,
      comparison_df
    )


  # ──────────────────────────────────────────────────────────
  # Print table
  # ──────────────────────────────────────────────────────────

  full_df %>%

    kable(

      caption =
        caption_text,

      col.names =
        c(
          "Parameter",

          "b", "SE", "p",
          "b", "SE", "p",
          "b", "SE", "p",
          "b", "SE", "p",
          "b", "SE", "p"
        ),

      align =
        c(
          "l",
          rep(
            "r",
            15
          )
        ),

      escape = FALSE
    ) %>%

    kable_styling(

      full_width = TRUE,

      font_size = 11,

      bootstrap_options =
        c(
          "condensed",
          "bordered"
        )
    ) %>%

    # THIS WAS THE MAIN ALIGNMENT ERROR
add_header_above(
  c(
```

```
    " " = 2,
    "Model 0" = 3,
    "Model 1" = 3,
    "Model 2" = 3,
    "Model 3" = 3,
    "Model 4" = 3
  )
    ) %>%

    footnote(

      general =
        footnote_text,

      general_title =
        "Note.",

      footnote_as_chunk =
        TRUE
    )
}
```

```
# ── INTEGRITY TABLE ──────────────────────────────────────

make_stepup_table(

  m0 = M0_Integrity,
  m1 = T1,
  m2 = T2,
  m3 = T3,
  m4 = T4,

  caption_text =
    "Table 8. Step-Up MLM Results for Integrity Ratings",

  footnote_text =
    "Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate =
0, and major = 1. Model comparison χ² values are likelihood-ratio tests comparing each model with the preceding model. * p <
.05. ** p < .01. *** p < .001."
)
```

Table 8. Step-Up MLM Results for Integrity Ratings

		Model 0			Model 1			Model 2			Model 3			Model 4		
	Parameter	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**
(Intercept)	Intercept	3.13	0.51	< .001***	3.13	0.18	< .001***	3.13	0.5	< .001***	3.13	0.15	< .001***	3.13	0.06	< .001***
role_ec	Role				1.18	0.18	< .001***				1.18	0.15	< .001***	1.18	0.06	< .001***
severity_num	QRP Severity							-0.29	0.61	0.655	-0.29	0.18	0.160	-0.29	0.07	0.006**
role_ec:severity_num	Role × QRP Severity													0.41	0.07	0.001**
responseid	Participant variance	0.041			0.041			0.041			0.041			0.041		
vignette_id	Vignette variance	1.565			0.185			1.509			0.129			0.018		
Residual	Residual variance	0.375			0.375			0.375			0.375			0.376		
1	ICC: Participant	0.019			0.052			0.019			0.055			0.065		
2	ICC: Vignette	0.719			0.232			0.712			0.175			0.029		
11	AIC	5826.3			5815.5			5828.1			5815.3			5805.8		
21	BIC	5861.6			5856.7			5869.3			5862.5			5858.9		
12	Model comparison χ^2	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***
22	Model comparison df	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***
3	Model comparison p	—			12.8	1	< .001***	0	0		14.72	1	< .001***	11.52	1	< .001***

Note. Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. Model comparison χ^2 values are likelihood-ratio tests comparing each model with the preceding model. * p < .05. ** p < .01. *** p < .001.

```
# ── LIKEABILITY TABLE ──────────────────────────────────

make_stepup_table(

  m0 = M0_like,
  m1 = L1,
  m2 = L2,
  m3 = L3,
  m4 = L4,

  caption_text =
    "Table 9. Step-Up MLM Results for Likeability Ratings",

  footnote_text =
    "Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate =
0, and major = 1. Model comparison χ² values are likelihood-ratio tests comparing each model with the preceding model. * p <
.05. ** p < .01. *** p < .001."
)
```

Table 9. Step-Up MLM Results for Likeability Ratings

		Model 0			Model 1			Model 2			Model 3			Model 4		
	Parameter	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**	**b**	**SE**	**p**
(Intercept)	Intercept	3.14	0.3	< .001***	3.14	0.13	< .001***	3.14	0.29	< .001***	3.14	0.12	< .001***	3.14	0.03	< .001***
role_ec	Role				0.67	0.13	0.002**				0.67	0.12	0.001**	0.67	0.04	< .001***
severity_num	QRP Severity							-0.16	0.36	0.665	-0.16	0.14	0.290	-0.16	0.03	0.003**
role_ec:severity_num	Role × QRP Severity													0.34	0.04	< .001***
responseid	Participant variance	0.077			0.077			0.077			0.077			0.077		
vignette_id	Vignette variance	0.538			0.098			0.52			0.08			0.004		
Residual	Residual variance	0.359			0.359			0.359			0.359			0.359		
1	ICC: Participant	0.06			0.092			0.061			0.093			0.103		
2	ICC: Vignette	0.42			0.117			0.412			0.097			0.005		
11	AIC	6191.7			6183.5			6193.5			6184.5			6169.5		
21	BIC	6268.3			6266			6276			6272.9			6263.8		
12	Model comparison χ^2	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***
22	Model comparison df	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***
3	Model comparison p	—			10.2	1	0.001**	0	0		10.99	1	< .001***	16.97	1	< .001***

Note. Role was effect coded as inspector = 1 and investigator = −1. QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. Model comparison χ^2 values are likelihood-ratio tests comparing each model with the preceding model. * p < .05. ** p < .01. *** p < .001.

```r
# ──────────────────────────────────────────────────────────────
# TABLE 10: PERCEIVED TRUENESS OF FINDING
# ──────────────────────────────────────────────────────────────

# Fixed effects
ft1 <- summary(FT1)$coefficients

# Random effects
ft1_vc <- as.data.frame(
  lme4::VarCorr(FT1)
)

# Extract variance components
participant_var <- ft1_vc$vcov[
  ft1_vc$grp == "responseid"
]

vignette_var <- ft1_vc$vcov[
  ft1_vc$grp == "severity"
]

residual_var <- ft1_vc$vcov[
  ft1_vc$grp == "Residual"
]

# Total variance
total_var <-
  participant_var +
  vignette_var +
  residual_var

# ICCs
participant_icc <-
  participant_var / total_var

vignette_icc <-
  vignette_var / total_var


# ──────────────────────────────────────────────────────────────
# Create table
# ──────────────────────────────────────────────────────────────

trueness_df <- data.frame(

  Parameter = c(
    "Intercept",
    "QRP Severity",
    "Participant variance",
    "Vignette variance",
    "Residual variance",
    "ICC: Participant",
    "ICC: Vignette",
    "AIC",
    "BIC"
  ),

  b = c(
    sprintf(
      "%.2f",
      ft1["(Intercept)", "Estimate"]
    ),

    sprintf(
      "%.2f",
      ft1["severity_num", "Estimate"]
    ),

    sprintf(
      "%.3f",
      participant_var
    ),

    sprintf(
      "%.3f",
      vignette_var
    ),

    sprintf(
      "%.3f",
      residual_var
    ),

    sprintf(
      "%.3f",
      participant_icc
```

```
    ),

    sprintf(
      "%.3f",
      vignette_icc
    ),

    sprintf(
      "%.1f",
      AIC(FT1)
    ),

    sprintf(
      "%.1f",
      BIC(FT1)
    )
  ),

  SE = c(
    sprintf(
      "%.2f",
      ft1["(Intercept)", "Std. Error"]
    ),

    sprintf(
      "%.2f",
      ft1["severity_num", "Std. Error"]
    ),

    "", "", "", "", "", "", ""
  ),

  p = c(
    fmt_p(
      ft1["(Intercept)", "Pr(>|t|)"]
    ),

    fmt_p(
      ft1["severity_num", "Pr(>|t|)"]
    ),

    "", "", "", "", "", "", ""
  ),

  stringsAsFactors = FALSE
)


# ──────────────────────────────────────────────────────────
# Print Table 10
# ──────────────────────────────────────────────────────────

trueness_df %>%

  kable(
    caption =
      "Table 10. MLM Results for Perceived Trueness of Finding",

    col.names = c(
      "Parameter",
      "b",
      "SE",
      "p"
    ),

    align = c(
      "l",
      "r",
      "r",
      "r"
    ),

    escape = FALSE
  ) %>%

  kable_styling(
    full_width = FALSE,
    font_size = 11,
    bootstrap_options =
      c(
        "condensed",
        "bordered"
      )
  ) %>%

  footnote(
```

```
    general =
      "Trueness was centered at 3 (neutral). QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. * p <
.05. ** p < .01. *** p < .001.",

    general_title =
      "Note.",

    footnote_as_chunk =
      TRUE
  )
```

Table 10. MLM Results for Perceived Trueness of Finding

Parameter	b	SE	p
Intercept	-0.88	0.11	0.003**
QRP Severity	-0.72	0.13	0.012*
Participant variance	0.167		
Vignette variance	0.033		
Residual variance	0.652		
ICC: Participant	0.196		
ICC: Vignette	0.039		
AIC	3503.7		
BIC	3529.7		

Note. Trueness was centered at 3 (neutral). QRP Severity was effect coded as minor = −1, moderate = 0, and major = 1. * p < .05. ** p < .01. *** p < .001.

```
cat("\n--- Integrity random effects ---\n")
```

```
##
## --- Integrity random effects ---
```

```
print(get_variance_components(T4))
```

```
##           grp Variance    SD
## 1  responseid    0.041 0.202
## 2  responseid    0.168 0.410
## 3  responseid    0.026 0.316
## 4 vignette_id    0.018 0.135
## 5    Residual    0.376 0.613
```

```
print(get_random_intercept_icc(T4))
```

```
##           grp   ICC
## 1  responseid 0.049
## 2  responseid 0.031
## 3 vignette_id 0.022
## 4    Residual 0.449
```

```
cat("\n--- Likeability random effects ---\n")
```

```
##
## --- Likeability random effects ---
```

```
print(get_variance_components(L4))
```

```
##            grp Variance     SD
## 1   responseid    0.077  0.277
## 2   responseid    0.244  0.494
## 3   responseid    0.006  0.074
## 4   responseid    0.048  0.218
## 5   responseid    0.036  0.260
## 6   responseid   -0.011 -0.532
## 7   responseid   -0.002 -0.025
## 8   responseid   -0.035 -0.943
## 9   responseid    0.024  0.226
## 10  responseid   -0.005 -0.331
## 11 vignette_id    0.004  0.064
## 12    Residual    0.359  0.599
```

```
print(get_random_intercept_icc(L4))
```

```
##           grp    ICC
## 1  responseid  0.094
## 2  responseid  0.043
## 3  responseid -0.013
## 4  responseid -0.002
## 5 vignette_id  0.005
## 6    Residual  0.437
```

```
cat("\n--- Trueness random effects ---\n")
```

```
##
## --- Trueness random effects ---
```

```
print(get_variance_components(FT1))
```

```
##          grp Variance    SD
## 1 responseid    0.167 0.409
## 2   severity    0.033 0.182
## 3   Residual    0.652 0.808
```

```
print(get_random_intercept_icc(FT1))
```

```
##          grp   ICC
## 1 responseid 0.111
## 2   severity 0.022
## 3   Residual 0.434
```